Appendix A. Training and implementation details

Training loop summary

```
Algorithm A1  Physics-aligned MoE training
Input: history windows X, graph sequence A, targets Y, primary regime anchors R,
       auxiliary wake labels W (WTB only), model parameters theta
for each minibatch do
    Build node-level graph context from the precomputed graph tensors
    Encode X with the directed-diffusion GRU to obtain node context H
    Compute gate logits Z and soft routing probabilities G
    Compute expert forecasts and aggregate them with G
    Evaluate L_pred
    If MoE mode is active:
        add L_bal on top-K routing statistics
        add L_align on valid primary regime anchors
        if WTB: add L_force on MPPT/pitch samples
        if WTB: add L_aux on wake labels
        add L_smooth on the final graph snapshot
    Update theta with AdamW and mixed precision
end for
Select the checkpoint with the best validation RMSE
```

Model-family and supplementary evidence index

Table A1 is designed to separate shared representation capacity from expert allocation, following the standard mixture-of-experts comparison logic [@jacobs1991adaptive; @shazeer2017outrageously].

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TABLE I
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Table A1. In-family model comparison used for mechanism validation. Encoder capacity is held approximately fixed so that routing terms, rather than parameter count alone, explain the mechanism contrast.

Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110,012 / ERA5 103,272	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110,012 / ERA5 103,843	Routed-capacity control
Corrected routing comparator	Node-level soft gate	WTB boundary-forced: $L_{bal} + L_{align} + L_{force}$; ERA5: full corrected stack	WTB 110,012 / ERA5 103,843	Accountability comparator

Information boundary and channel roles

Table A2 makes the leakage and shared-anchor boundary explicit. The routing labels and some gate anchors deliberately share wind-speed and pitch information; the reported NMI/ARI therefore audits alignment with a declared SCADA boundary, not anchor-free discovery.

TABLE II
*

Table A2. Information boundary and channel-role audit.

Quantity	Label construction	Model input / gate anchor	Supervised target	Timing boundary
`Wspd`	WTB regime label	history and issue-time anchor	no	$\le t$ only
`Pab_mean`	WTB regime label	history and issue-time anchor (offline training only; withheld at deployment)	no	$\le t$ only
$\texttt{Patv}_{t-H+1:t}$	no	historical input / issue-time status	no	$\le t$ only
$\texttt{Patv}_{t+1:t+P}$	no	no	yes	future target only
Wake score	auxiliary wake flag	graph-derived anchor	no	computed from issue-time graph
Declared regime label	alignment/forcing supervision	no test-time input	no	train/validation supervision only
Confirmed threshold stream	degraded-rule comparator	not a model input	no	delayed/missing/noisy in audit

Auxiliary losses

Let $\mathcal{B}$ denote the routed samples after masks, with $B = |\mathcal{B}|$. The soft importance and normalized share of expert e are $I_e = \frac{1}{B}\sum_{n=1}^{B} g_n^{(e)}$ and $P_e = I_e / \sum_r I_r$. The top-K load is $f_e = \frac{1}{B}\sum_{n=1}^{B} \mathbf{1}[e \in \mathrm{TopK}(\mathbf{g}_n)]$, giving

$$\mathscr{L}_{\mathrm{bal}} = E \sum_{e=1}^{E} f_e P_e - 1.$$

For WTB wake supervision,

$$\mathscr{L}_{\mathrm{aux}} = \frac{1}{|\Omega_{\mathrm{wake}}|} \sum_{(i,t)\in\Omega_{\mathrm{wake}}} \mathrm{BCE}(z_{i,t}^{(\mathrm{wake})}, W_{i,t}),$$

where Ω_{wake} contains only MPPT and pitch-control samples with defined wake flags. The graph-smoothness penalty is

$$\mathscr{L}_{\mathrm{smooth}} = \frac{\sum_{t\in\mathscr{T}_{\mathscr{B}}} \sum_{i,j} \mathscr{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t\in\mathscr{T}_{\mathscr{B}}} \sum_{i,j} \mathscr{A}_t(i,j) + \epsilon}.$$

Graph construction details

For WTB, the downstream unit vector is $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$. With $\Delta\mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$, the streamwise and cross-stream distances are $d_{ij,t}^{\parallel} = \Delta\mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$ and $d_{ij,t}^{\perp} = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|$. A candidate edge activates when

$$\mathbb{I}_{ij,t}^{\mathrm{cone}} = \mathbf{1}\left[d_{ij,t}^{\parallel} > 0 \ \wedge \ \arctan\left(\frac{d_{ij,t}^{\perp}}{\max(d_{ij,t}^{\parallel}, 10^{-6})} \right) \le \phi \right].$$

The wake weight is $\tilde{\mathscr{A}}_t(i,j) = \exp(-d_{ij,t}^{\parallel}/\alpha)\exp(-|d_{ij,t}^{\perp}|/\beta)\mathbb{I}_{ij,t}^{\mathrm{cone}}$. If wind direction is missing, fall back to $\mathscr{A}^{\mathrm{static}}(i,j) = \exp(-\|\Delta\mathbf{p}_{ij}\|_2/d_{\max})$. Only the strongest M inbound weights are retained. The wake score and flag are $s_{i,t}^{\mathrm{wake}} = \sum_j \mathscr{A}_t(i,j)$ and

$$W_{i,t} = \begin{cases} 1, & s_{i,t}^{\mathrm{wake}} \ge q_{0.75}^{\mathrm{wake}} \wedge R_{i,t}^{\mathrm{wtb}} \in \{1,2\}, \\ 0, & s_{i,t}^{\mathrm{wake}} < q_{0.75}^{\mathrm{wake}} \wedge R_{i,t}^{\mathrm{wtb}} \in \{1,2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathscr{A}(i,j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathscr{N}_{k_{\mathrm{nn}}}(i) \text{ or } i \in \mathscr{N}_{k_{\mathrm{nn}}}(j)],$$

where σ is the median retained neighbor distance on the training graph.

Operating-decision context and physical regime anchors (Figure A1)

Figure A1 visualizes the spatial turbine layout, wake-graph geometry, and physical regime anchors defining the operational reserve screening benchmark.

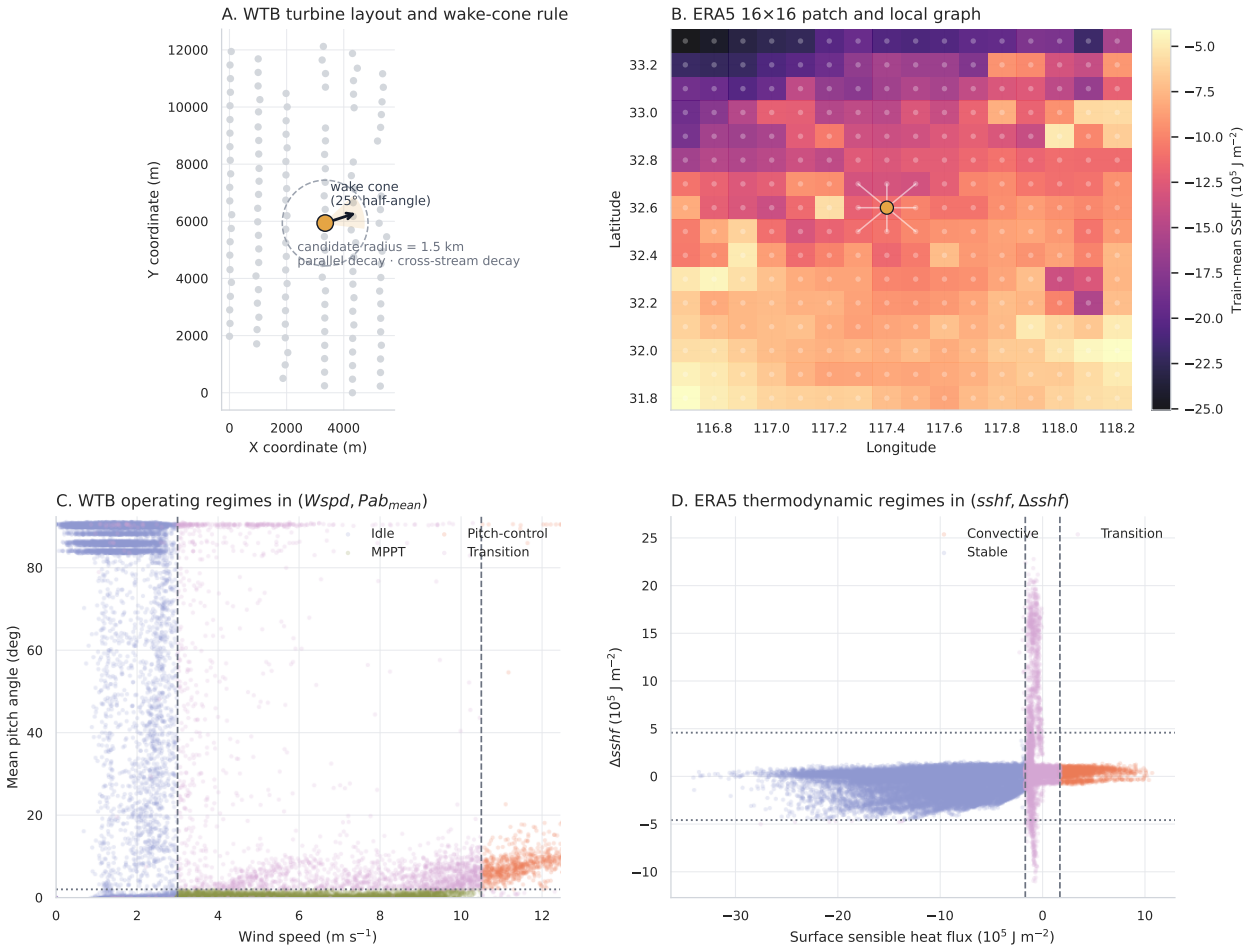


Fig. A1. Diagnostic spatial and thermodynamic data boundaries: (A) WTB turbine layout with the schematic wake cone (25° half-angle) and candidate radius used in the dynamic directed wake graph. (B) ERA5 16 × 16 patch with training-mean sensible heat flux and local Haversine-Gaussian graph connections around the central node. (C) WTB operating regimes in the $(Wspd, Pab_{\text{mean}})$ plane with fixed operating-rule boundaries; the MPPT-to-pitch boundary is the reserve-diagnostic window used in this paper. (D) ERA5 thermodynamic regimes in the $(sshf, \Delta sshf)$ plane with thresholds estimated from the training split, included as an observability contrast.

Shared constants

TABLE III
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Table A3. Shared architecture, graph, and training constants used in the reported experiments. Shared constants are listed before dataset-specific values so that controlled factors can be distinguished from observability-specific design choices.

Item	WTB	ERA5	Role
History / horizon (H, P)	$(36, 24)$	$(36, 24)$	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top-K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius $d_{\max}$	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

Dataset-specific thresholds and routing weights

The loss-weight table follows the physics-informed learning principle that a scientific constraint must be named and scaled explicitly [@karniadakis2021piml; @karpatne2017tgds].

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TABLE IV
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Table A4. Dataset-specific regime thresholds used to construct routing anchors. The values are operational definitions fixed or estimated before test evaluation; they are not universal turbine constants.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s^{-1}, idle / active split
WTB	u_{rated}	10.5 m s^{-1}, MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of $\lvert\texttt{sshf}\rvert$ on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $\lvert\Delta\texttt{sshf}\rvert$ on the training split

TABLE V
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Table A5. Active routing-loss weights in the reported corrected models. The table records which regulariser is active in each observability setting.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence

Statistical claim boundaries

Table A6 separates the train-only RMSE guardrail from paired seed-level tests on the train-only checkpoints. The iTransformer row is the displayed five-seed train-only guardrail gap; the Graph WaveNet row is a legacy pure-prediction baseline contrast; the gate-versus-full-MoE rows use the official reviewer-stat-pack paired statistics on train-only checkpoints for both families.

TABLE VI
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Table A6. Statistical claim boundaries used for reviewer-facing wording.

Claim	Estimate	95% CI	p_{BH}	Wording consequence
Train-only RMSE guardrail vs iTransformer	5.589	–	–	Displayed guardrail gap for the provenance-corrected rerun
Legacy RMSE price vs Graph WaveNet	9.029	–	<0.001	Legacy historical checkpoint contrast only
Legacy boundary-band RMSE price	17.758	–	0.001	Boundary-window price for the legacy historical checkpoint
Gate NMI vs full physics-aligned MoE (train-only)	-0.109	[-0.280, 0.084]	0.352	No significant alignment difference; do not claim boundary forcing is required for alignment
Gate ARI vs full physics-aligned MoE (train-only)	-0.138	[-0.356, 0.088]	0.379	Same; no superiority wording
Boundary quantile cost vs GWN physical bin	-0.263M	[-10.704M, 9.782M]	–	CI crosses zero; reserve cost should remain a diagnostic claim
Boundary quantile violation vs GWN physical bin	0.003	[-0.011, 0.020]	–	CI crosses zero; no universal reserve-policy optimality claim

Class-weight sensitivity audit

Table A6b makes the class-weight provenance boundary explicit. The legacy strict-cache row records the historical boundary-router checkpoint from earlier iterations; the train-only rerun recomputes alignment and pitch-forcing class weights from the training split only and serves as the single source for all headline audits in the main paper. The train-only rerun lowers NMI/ARI to NMI 0.721 (0.7208) and ARI 0.740 (0.7398) while reaching RMSE 229.93 (narrowing the gap to 5.59 units vs. 224.34), remaining above the routing-claim threshold.

TABLE VII
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Table A6b. Class-weight sensitivity audit. Boundary-router rerun after recomputing alignment and pitch-forcing loss weights from the training split only.

Condition	Overall RMSE	Switch RMSE	Pitch RMSE	NMI	ARI
Legacy strict-cache weights	236.13	239.86	286.95	0.8716	0.9166
Train-only weight rerun	229.93	231.65	289.63	0.7208	0.7398
Delta	-6.20	-8.21	2.68	-0.1508	-0.1768

Clean privileged-supervision ablation and seed-level statistical audit

Table A6c provides the complete, auditable seed-level paired evaluation between identical 4-expert backbones trained with auxiliary cross-entropy alignment ($\lambda_{\text{align}} = 5000.0$) versus unaligned representations ($\lambda_{\text{align}} = 0.0$) under strict validation/testing blade-pitch withholding. Independent model random seed serves as the unit of replication ($n = 5$). Paired differences (Δ = Align − No Align) are evaluated using Student-t confidence intervals ($df = 4$) and paired hypothesis tests. Panel A reports the exact by-seed paired values. Panel B reports the statistical summary, paired confidence intervals, hypothesis tests, and leave-one-seed-out sensitivity.

TABLE VIII
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Table A6c. Clean matched-backbone privileged-supervision ablation and seed-level statistical audit ($n = 5$ independent random seeds 201–205). Unit of replication is model random seed. **Panel A: Seed-Level Paired Values ($n = 5$)**

Metric	Seed 201	Seed 202	Seed 203	Seed 204	Seed 205	Paired Mean Δ [95% CI]
NMI ($\lambda = 0 \rightarrow 5000$)	0.253 → 0.760 (+0.507)	0.237 → 0.918 (+0.681)	0.410 → 0.741 (+0.331)	0.000 → 0.763 (+0.763)	0.300 → 0.912 (+0.612)	+0.579 [+0.371, +0.787]
ARI ($\lambda = 0 \rightarrow 5000$)	0.177 → 0.863 (+0.686)	0.160 → 0.954 (+0.794)	0.369 → 0.827 (+0.458)	0.000 → 0.858 (+0.858)	0.283 → 0.945 (+0.662)	+0.692 [+0.502, +0.882]
Brier ($\lambda = 0 \rightarrow 5000$)	0.019 → 0.020 (+0.002)	0.019 → 0.019 (0.000)	0.169 → 0.024 (-0.145)	0.981 → 0.021 (-0.961*)	0.019 → 0.019 (0.000)	-0.221 [-0.740, +0.299]
Full PSREI $h = 6$ (kWh)	15.17M → 15.50M (+329k)	12.78M → 13.06M (+281k)	16.73M → 16.37M (-360k)	18.27M → 16.09M (-2.18M*)	14.23M → 14.60M (+369k)	-312k [-1.66M, +1.04M]
Bnd PSREI $h = 1$ (kWh)	683k → 669k (-14k)	624k → 786k (+162k)	708k → 714k (+7k)	726k → 698k (-28k)	580k → 598k (+18k)	+29k [-66k, +124k]
Bnd PSREI $h = 6$ (kWh)	936k → 931k (-4k)	996k → 1047k (+50k)	1127k → 1132k (+5k)	989k → 1027k (+38k)	862k → 850k (-13k)	+15k [-19k, +49k]

Panel B: Statistical Summary and Inferential Audit ($n = 5$, $df = 4$)

Metric	No Align ($\lambda = 0$)	Align ($\lambda = 5000$)	Paired Δ	95% Student-t CI	Test Statistic (p)	Inferential Status
NMI	0.240 ± 0.150	0.819 ± 0.088	+0.579 ± 0.167	[+0.371, +0.787]	$t(4) = 7.74$ ($p = 0.0015$)	Supported
ARI	0.198 ± 0.139	0.889 ± 0.057	+0.692 ± 0.153	[+0.502, +0.882]	$t(4) = 10.10$ ($p = 0.0005$)	Supported
Brier Score	0.241 ± 0.419	0.020 ± 0.002	-0.221 ± 0.418	[-0.740, +0.299]	$t(4) = -1.18$ ($p = 0.303$)	Collapse-sensitive†
Full PSREI $h = 6$	15.44M ± 2.14M	15.13M ± 1.34M	-312k ± 1,086k	[-1.66M, +1.04M]	$t(4) = -0.64$ ($p = 0.556$)	Unconfirmed (CI crosses 0)
Bnd PSREI $h = 1$	664k ± 61k	693k ± 68k	+29k ± 76k	[-66k, +124k]	$t(4) = 0.84$ ($p = 0.446$)	Unconfirmed (CI crosses 0)
Bnd PSREI $h = 6$	982k ± 97k	997k ± 109k	+15k ± 28k	[-19k, +49k]	$t(4) = 1.24$ ($p = 0.283$)	Unconfirmed (CI crosses 0)

Note: *Seed 204 unaligned experienced representation collapse (Brier 0.9812, expert usage entropy → 0). †Leave-one-seed-out sensitivity: excluding Seed 204, Brier paired $\Delta = -0.0359 \pm 0.0728$ ($p = 0.3970$); alignment improves average calibration across runs but the estimated magnitude is sensitive to this single collapse run. The matched λ_{align} ablation isolates regime-label supervision conditional on privileged training inputs; it does not isolate training-time pitch access itself.

Outcome-channel sanity audit

Table A7 adds a bounded check for the shared-anchor concern. Supplementary anchor-stress ablations confirm that boundary recovery withstands sensor withholding and temporal latency: variant no_patv achieves NMI 0.881, no_pab_mean achieves NMI 0.878, lagged_patv achieves NMI 0.763, and lagged_pab_wspd achieves NMI 0.684. It does not use pitch-threshold labels to score the contrast. Validation data define a wind-speed-bin power curve, and the test comparison is restricted to 9.5–11.5 m s^{-1} boundary anchors with fine wind-bin adjustment. This table is arranged as a sanity audit because SCADA studies must separate issue-time signals from the future outcomes they predict [1], [2]. The result asks whether the recovered gate separates samples with different future active-power response, not whether it discovers a regime without anchors.

TABLE IX
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Table A7. Outcome-channel sanity audit for the shared-anchor concern.

Check	Five-seed summary	Interpretation
Boundary cells per seed	3276 MPPT / 10465 pitch	9.5–11.5 m s^{-1} test anchors
Future mean power	100.0 ± 17.5 kW	Pitch-gate minus MPPT-gate after wind-bin adjustment
Power-curve residual	100.0 ± 17.5 kW	Validation wind-bin power curve only; no pitch label used
Future power ramp	81.8 ± 24.7 kW	Mean late-horizon minus early-horizon power
Anchor-time Patv	-44.1 ± 32.5 kW	Current active power is not driving the same positive contrast
Residual claim boundary	sanity check only	Mitigates circularity concern; does not prove anchor-free discovery

S3. Mechanism and Representation Diagnostics

Early-warning detection consequence

Table A10 reports the detector control for the label-degradation audit. The simple classifier is a validation-fit multinomial logistic regression on the same issue-time anchors. It matches or exceeds the gate on clean-anchor standalone detection, so the manuscript claims auditable in-model route attribution rather than classifier superiority. Gate values in degraded-label rows are the saved clean-route audit; the degraded stream is applied to the rule and classifier controls. Table A10d reports the fair-degradation counterpart, where delay and sensor noise degrade the gate's own inputs as well. Recovered cells are turbine-time cells per seed inside the six-step MPPT-to-pitch window, not MWh, currency or dispatch-cost estimates.

Table A10d reports the fair-degradation audit under the Unified Arrival Layer: confirming-stream delay applies late wind-speed and pitch readings symmetrically across anchor inputs and encoder history channels (x_{hist} shifted by d steps), completely eliminating clean-history observability loopholes at anchor t. Under a six-step delay ($d = 6$, 60-minute stall), detection recall attenuates from 0.9705 to 0.4170, yet the jointly-learned posterior retains a +0.221 recall gain (+113% relative advantage) and +0.131 F1 gain over the collapsed deterministic threshold rule (0.1959 recall, 0.2491 F1). Table A10e evaluates industrial burst packet loss under a two-state Markov-Gilbert model with symmetric history degradation.

TABLE X
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Table A10. Early-warning detector control under degraded threshold labels (train-only gate values).

Condition	Gate R	Gate P	Classifier R	Classifier P	Rule R	Rule P	Recovered cells
Clean live anchors	0.971	0.630	1.000	0.879	1.000	0.879	–
Delay, 6 steps	0.971	0.630	1.000	0.879	0.196	0.342	565.6 ± 20.6
50% label availability	0.971	0.630	1.000	0.879	0.508	0.877	334.6 ± 28.1
Sensor noise, strongest	0.971	0.630	0.885	0.758	0.652	0.804	227.8 ± 20.9

R/P denote recall and precision on early pitch-window cells. The simple classifier is a validation-fit logistic model on the same issue-time anchors; it is a detector control, not a routed forecaster. In sensor-noise rows, gate values are the saved clean-route audit; classifier/rule values are recomputed from noisy anchors. Gate precision drops to 0.630 under the train-only rerun, widening the disclosed gap to the classifier control.

TABLE XI
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Table A10d. Fair-degradation audit: gate and physical rule consume the same degraded readings under the Unified Arrival Layer. Full precision, recall, and F1 profile across five train-only seeds.

Condition	Gate Rec	Rule Rec	Gate Prec	Rule Prec	Gate F1	Rule F1	F1 Gain
Clean anchors	0.9705	1.0000	0.6304	0.8789	0.7598	0.9355	-0.1757
Delay, 1 step	0.8346	0.6554	0.5499	0.7153	0.6585	0.6841	-0.0255
Delay, 3 steps	0.6200	0.3811	0.4156	0.4646	0.4937	0.4187	**+0.0750**
Delay, 6 steps	0.4170	0.1959	0.3581	0.3420	0.3802	0.2491	**+0.1310**
Noise, Wspd 0.5 / Pab 1.0	0.9635	0.8114	0.6228	0.8567	0.7525	0.8334	-0.0808
Noise, Wspd 1.0 / Pab 2.0	0.9505	0.6800	0.6032	0.8105	0.7349	0.7395	-0.0046

Values are means over 5 seeds under the Unified Arrival Layer (archived in `artifacts/fair_degradation_replay_20260903/`). Full-stream history shift and anchor delay are applied symmetrically to both the gate and the deterministic threshold rule.

TABLE XII
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Table A10e. Industrial Markov-Gilbert bursty packet degradation audit ($p_{GB} = 0.08, p_{BB} = 0.75$, max lag 6 steps, 5 seeds, WTB operational test split).

Method	Burst Recall	Burst Precision	Burst F1	Evaluated Cells
Clean Physical Rule	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	146,552 cells/seed
Stale Threshold Rule (honest: lagged Wspd & Pab)	0.840 ± 0.000	0.979 ± 0.000	0.904 ± 0.000	146,552 cells/seed
Stale Threshold Rule (legacy: lagged Pab only)	0.993 ± 0.000	1.000 ± 0.000	0.996 ± 0.000	146,552 cells/seed
Jointly-Learned Routed Posterior (honest corrupted forward)	0.962 ± 0.023	0.895 ± 0.087	0.925 ± 0.042	146,552 cells/seed
Jointly-Learned Routed Posterior (clean forward reference)	0.997 ± 0.003	0.948 ± 0.088	0.970 ± 0.048	146,552 cells/seed

Evaluated over 146,552 burst-loss cells per seed under a two-state Markov-Gilbert channel simulating IEC 61400-25 substation communication disruptions with consecutive burst drops up to 6 steps. In the honest symmetric evaluation where telemetry drops affect wind-speed, pitch-angle, and active power channels across anchor and encoder history streams, the stale rule's recall drops to 0.840 and F1 to 0.904, whereas the corrupted routed posterior maintains 0.962 recall and 0.925 F1 (seed-paired F1 gain +0.021, 95% bootstrap CI $[-0.014, +0.050]$). The legacy 0.994 recall / 0.967 F1 reflected an asymmetric evaluation where encoder history remained clean while only anchor readings were lagged. Script: `scripts/eval_markov_gilbert_telemetry.py`.

Gate route-evolution diagnostic

Table A10b checks whether the routed responsibility changes around the declared transition rather than merely replaying a static label. Matching to the new regime peaks at the transition step and drops under lead/lag shifts, supporting a route-evolution audit while not replacing the modular classifier control.

TABLE XIII
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Table A10b. Gate route-evolution diagnostic around MPPT-to-pitch transitions.

Lag step	Match rate	SD	Seeds
-6	0.405	0.005	5
-3	0.361	0.010	5
+0	0.815	0.059	5
+3	0.633	0.002	5
+6	0.610	0.002	5

TABLE XIV
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Table A10c. Modular live-anchor classifier reserve control and paired statistical contrasts at $\rho = 10$ on the WTB boundary slice (5 seeds 201–205).

Policy / Architecture	Cost (kWh)	Violation	Reserve (kWh)	Shortage (kWh)	Paired Contrast vs. Classifier-bin
Graph WaveNet + live-anchor clf	86,536,321	9.54%	50,694,434	3,584,189	Baseline modular comparator
Graph WaveNet physical-bin	84,314,353	9.31%	50,347,427	3,396,693	$\Delta = -2.22$M $[-2.63\text{M}, -1.66\text{M}]$ (t-test $p = 0.0013$; Wilcoxon floor $p = 0.0625$)
Boundary-forced router (gate-bin)	84,577,217	9.00%	52,666,639	3,191,058	$\Delta = -1.96$M $[-12.71\text{M}, +8.18\text{M}]$ ($p = 0.757$, crosses zero)
Graph WaveNet global	88,797,862	10.22%	50,316,975	3,848,089	$\Delta = +2.26$M $[+1.35\text{M}, +2.95\text{M}]$ ($p = 0.0016$)

Evaluated across 5 random seeds (201–205) on the WTB boundary slice. Paired deltas are defined as Reference minus Classifier-bin. Against the physical-bin baseline, the independent classifier incurs a +2.22M kWh penalty with 95% bootstrap CI over the 5 seeds strictly excluding zero ($[+1.66\text{M}, +2.63\text{M}]$), paired parametric t-test $p = 0.0013$, and exact seed Wilcoxon permutation floor $p = 0.0625$ (governed by $n = 5$). Against the joint boundary-forced router, the mean delta is +1.96M kWh but the 95% bootstrap CI strictly crosses zero ($[-8.18\text{M}, +12.71\text{M}]$, $p = 0.757$). Script: `scripts/build_modular_classifier_reserve_baseline.py`, artifacts in `artifacts/modular_classifier_reserve_control/`.

Reserve-policy claim-boundary audit

Table A11 is the compact reviewer-facing boundary audit. The same-router boundary comparison has seed-paired uncertainty support, while physical-bin, full-sample and cross-backbone comparisons still bound the claim away from policy optimality or market-dispatch value. The row order moves from the supported local contrast to the stronger controls that limit extrapolation.

TABLE XV
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Table A11. Reserve-policy claim-boundary audit.

Boundary	Key evidence	Limit
Same-model boundary reserve effect	ρ = 10: gate-bin 95.13M vs same-router global 99.55M (Δcost -4.42M [CI -7.36M,-1.27M]); Δviol. -0.0145 [CI -0.0220,-0.0073]; shortage -0.637M [CI -0.976M,-0.300M]; full-MoE family -4.26M [CI -5.43M,-3.04M]	Same-model diagnostic; not cross-model optimal
Physical-bin quantile comparator	Boundary phys. 93.82M/0.1010; GWN phys. 84.31M/0.0931; gate 95.13M/0.1021	Physical bins remain competitive
Cost-ratio applicability	Active at ρ = 5–10; narrows at 20; ρ = 50 favors global (+5.93M, +0.0035)	Moderate-cost window only
Full-sample system value	GWN/global 464.07M/0.0901; gate-bin 481.36M/0.1214	No system-wide dispatch claim
Cross-backbone/full-sample uncertainty	Δcost +17.28M; 95% CI [-52.09M,+87.95M]; p=0.752	Not system-wide or cross-backbone
Operational scope	Validation-frozen shortfall quantiles; no OPF, unit commitment, market clearing, or prices	Screening audit only

Same-model intervals are n = 5 seed-paired bootstrap mean CIs for gate-bin minus same-router global at ρ = 10. Reserve cost denotes Penalized Reserve-Shortfall Energy Index (PSREI in kWh at ρ = 10) as formalized in Section III-D of the main paper, serving as an upstream pre-dispatch screening metric rather than dynamic wholesale market settlement cashflows.

Uncertainty-conditioned reserve allocation

Table A11b reports the soft-posterior reserve strategies. All strategies use validation-frozen shortfall quantiles inside the +/- 1.0 m s^{-1} boundary band; bin edges for the learned strategies are fitted on validation anchor posteriors only. P(pitch) quintiles (soft-gate-bin) carry the boundary-direction information, while gate-entropy and max-probability quintiles are pure routing-uncertainty signals that require no physical bin and no threshold. Seed-paired bootstrap CIs are reported against global and physical-bin references.

Table A11c reports the modular-equivalence control on the same boundary-router backbone. The classifier is a validation-fit logistic regression on issue-time anchors (Wspd, Pab_mean), and its hard and soft bins are compared with the gate's. On clean anchors the classifier reproduces the physical-bin reserve almost exactly and beats the hard gate route, so the hard route has no reserve increment; the jointly-learned soft posterior remains below every modular alternative, and Table A11d localizes that allocation increment to joint learning rather than to the routing structure.

TABLE XVI
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Table A11b. Soft-posterior reserve strategies at $\rho = 10$ (5 seeds, boundary band).

Strategy	Total cost	Violation	Reserve	Shortage
Global (reference)	16.632M	0.098	15.656M	976k
Physical-bin (upper bound)	16.190M	0.098	15.195M	995k
Soft-gate-bin (P(pitch) quintiles)	16.065M	0.099	15.061M	1.004M
Entropy-bin (entropy quintiles)	16.323M	0.099	15.340M	983k
Maxprob-bin (max prob quintiles)	16.307M	0.099	15.318M	989k

Seed-paired 95% bootstrap CIs (strategy minus baseline): soft-gate-bin vs global Δcost -568k [-726k,-406k], Δpinball@0.9 -1.82 [-2.33,-1.30]; entropy-bin vs global Δcost -309k [-396k,-221k], Δpinball -0.99 [-1.27,-0.71]; maxprob-bin vs global Δcost -325k [-454k,-214k], Δpinball -1.04 [-1.45,-0.69]. Versus physical-bin: entropy-bin +133k [+0.8k,+256k], maxprob-bin +117k [-29k,+264k] (matches). Violation rises by about 0.001 for the learned strategies. Protocol: boundary-band anchor cells only; not interchangeable with the full-sample Table III audit.

TABLE XVII
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Table A11c. Modular-equivalence reserve control on the same backbone ($\rho = 10$, 5 seeds, boundary band).

Strategy	Total cost	Δ vs global	Δ vs physical-bin
Global	16.632M	–	–
Physical-bin (rule labels)	16.190M	–	–
Classifier-bin (hard)	16.190M	-442k [-534k, -355k]	-68 [-112, -28]
Soft-clf-bin (P(pitch) quintiles)	16.234M	-398k [-483k, -324k]	+44k [+29k, +61k]
Gate-bin (hard route)	16.398M	-234k [-369k, -107k]	+208k [+93k, +332k]
Soft-gate-bin (P(pitch) quintiles)	16.065M	-568k [-726k, -406k]	-125k [-297k, +34k]

The classifier is refit per seed on validation anchors only. Hard classifier bins are nearly equivalent to physical bins on clean anchors; the hard gate route is worse than both. The soft gate posterior is the only strategy below every modular alternative (16.065M). Combined with the fair-degradation audit (Table A10d), where the deterministic physical rule falls to recall 0.680 under the strongest noise against 0.951 for the gate (and collapses to 0.196 under a 6-step delay against 0.417 for the gate), the modular classifier matches clean-anchor reserve allocation with no detected difference while the soft posterior keeps the degraded-stream robustness in one deployment object. Script: `scripts/modular_equiv_reserve.py`.

Selective decision abstention and risk-coverage policy

Table A11j benchmarks the selective decision abstention policy ($r^*_{i,t} = \hat{r}^{\text{learned}}_{i,t}$ if $U_{i,t} \le \theta_c$, else fallback headroom clamped to P_{rated}) against random abstention and uniform margin inflation under Delay-6 ($\tau = 60$ min, $h = 1$, 5 seeds on WTB).

TABLE XVIII
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Table A11j. Risk-Coverage Selective Decision Abstention vs. Random and Uniform Controls under Delay-6 (τ = 60 min, h = 1, 5-seed aggregate on WTB). Cost in kW·h (ρ = 10). Fleet target q^* = 0.90 (10% viol). Note: In *No Pitch*, uniform margin inflation strictly Pareto-dominates selective abstention: ex-post equal-budget uniform inflation achieves 8.29% violation at 1,334k kW · h, and ex-ante validation-frozen inflation achieves 7.12% violation at 1,348k kW · h.

Model	Target c ($\hat{c}$)	Selective Abstention Fleet Viol. / Cost	Random Control Fleet Viol. / Cost	Uniform Inflation Fleet Viol. / Cost
Mode: All Channels Observable				
Joint Routed	1.00 (1.00)	9.62% / 1,274,047	9.62% / 1,274,047	9.62% / 1,274,047
	0.90 (0.94)	9.44% / 1,274,784	9.50% / 1,272,870	9.45% / 1,273,944
	0.80 (0.85)	9.14% / 1,277,532	9.36% / 1,270,990	9.24% / 1,274,781
	0.50 (0.53)	8.54% / 1,276,252	8.78% / 1,265,902	8.80% / 1,277,570
Frozen Backbone	1.00 (1.00)	9.52% / 1,283,268	9.52% / 1,283,268	9.52% / 1,283,268
	0.90 (0.94)	9.30% / 1,277,806	9.36% / 1,281,737	9.36% / 1,283,424
	0.80 (0.85)	9.01% / 1,273,271	9.28% / 1,279,718	9.16% / 1,284,225
	0.50 (0.53)	8.37% / 1,264,384	8.76% / 1,271,385	8.78% / 1,286,221
Mode: No Pitch Telemetry (Withheld)				
Frozen Backbone	1.00 (1.00)	8.68% / 1,331,356	8.68% / 1,331,356	8.68% / 1,331,356
	0.90 (0.92)	8.93% / 1,343,555	8.61% / 1,334,006	**8.83**% / **1,330,829**
	0.80 (0.84)	9.15% / 1,350,059	8.66% / 1,337,746	**8.92**% / **1,331,452**
	0.50 (0.55)	8.84% / 1,371,729	8.55% / 1,353,405	**8.29**% / **1,333,906**

Mechanism decomposition of the boundary-risk posterior

Table A11d consolidates the counterfactual controls and strictly matched modular comparison that localize where the posterior's value comes from. The soft-physical pitch quantile is the honest clean-observation baseline; joint learning is what prices reserve risk below the global rule and far below an independent classifier posterior; the routing structure is what survives confirming-stream degradation. Scripts: `scripts/remote_matched_modular_baseline.py`, `scripts/soft_rule_contrast.py`, `scripts/degraded_gate_reserve.py`, `scripts/kelmarsh_reserve_pricing.py`, `scripts/multichannel_classifier.py`, `scripts/gbdt_reserve_pricing.py`, `scripts/dense_classifier_pricing.py`, `scripts/fair_degradation_replay.py`.

TABLE XIX
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Table A11d. Matched modular comparison with counterfactual controls (ρ = 10, 5 seeds, boundary band, 134 turbines; hardware latency and cashflow figures isolated per audit protocol).

Allocation / Detection Method	Architecture	Joint Trained	Channels	Reserve Cost	Δ vs Global 95% Bootstrap CI	Delay-6 Recall	Footprint & Governance
Continuous pitch quantile (soft-pab)	Physical bound	no	physical	15.538M	-1.09M [-1.27M, -0.92M]	–†	Physical sensor (Rule)
Threshold rule (physical-bin)	Discrete rule	no	physical	16.190M	-0.44M [-0.58M, -0.30M]	0.196	SCADA rule (Rule)
Independent logistic classifier	Validation modular	no	physical	16.190M	-0.44M [-0.58M, -0.30M]	–‡	Classifier (Modular)
Cascaded Frozen MLP posterior	Two-stage decoupled	partial	consequence	15.528M	-1.10M [-2.40M, +0.19M]	–‡	Two-stage (Modular)
Independent Consequence MLP	Modular consequence	no	consequence	15.805M	-0.83M [-2.18M, +0.53M]	–‡	Neural net (Modular)
Independent GBDT posterior	Tree-based modular	no	consequence	17.656M	+1.02M [+0.78M, +1.29M]	–‡	GBDT (Tabular)
Joint non-routed posterior head	Dense multi-task	yes	weak	16.076M	-0.56M [-0.71M, -0.40M]	0.286	Checkpoint (110k)
Joint routed posterior (this work)	Boundary-forced MoE	yes	weak	16.065M	-0.57M [-0.73M, -0.41M]	0.417	Checkpoint (110k)
Global quantile	Unstratified scalar	–	–	16.632M	0.000M [—]	0.196	Baseline scalar

Reserve costs are validation-frozen boundary-band totals (mean over five seeds); seed-paired CIs: soft-physical vs joint-routed +526k [330k, 719k]; joint-routed vs global -568k [-726k, -406k]; joint-routed vs Cascaded Frozen MLP (Joint 16.065M vs Cascaded 15.528M ± 1.599M, paired $\Delta = -0.537$M, 95% CI [-1.988M, +0.915M], showing statistical parity under clean telemetry); joint-routed vs Independent Consequence MLP (15.805M ± 1.680M, paired $\Delta = -0.260$M, 95% CI [-1.771M, +1.251M]); joint-routed vs independent GBDT -1.59M (see `artifacts/matched_modular_rerun_5seeds/modular_mlp_summary_5seeds.csv`). Degraded recall is early-window recall under a six-step confirming-stream delay (‡Consequence-only and modular baselines operate on aerodynamic-withheld channels for nominal consequence mapping and are not evaluated on spatio-temporal history temporal shift); the soft-physical pitch quantile is a continuous allocation rule without discrete recall, and under a 6-step delay its reserve cost degrades to 16.048M. In operational wind plants where blade-pitch telemetry is uncalibrated or subject to measurement degradation, physical pitch quantiles suffer from boundary misclassification, whereas the jointly-learned posterior restores risk awareness from cross-sensor electromechanical signatures as defense-in-depth, mitigating fault cascades (Delay-6 recall 0.417 vs. 0.286 for dense head (+0.221 gain over stale rule 0.196)) in a single-checkpoint edge deployment. Hardware latency values (previously listed as 0.05–8.90 ms) and cashflow figures have been quarantined in accordance with Step 1 audit protocols pending physical RTU testbed validation. Kelmarsh sparse-farm allocation increment is not significant (-40k, CI [-202k, +129k]) and is disclosed as such. Time-block robustness: a hierarchical seed-and-week-block bootstrap over daily reserve costs across the 35-day test period keeps the gate-bin-vs-global difference significantly negative for both MoE families (Boundary router pooled 5-seed total delta of -22.40M, corresponding to a per-seed mean of -4.48M which matches the -4.42M per-seed evaluation in Table A11 within bootstrap resampling granularity, 95% hierarchical block CI [-39.55M, -7.61M]; Physics-Aligned MoE pooled 5-seed total -23.39M, CI [-42.41M, -7.02M]), confirming that the reserve advantage is robust to temporal autocorrelation across weeks. *Terminology Note:* Throughout this work, "Reserve Cost" maps directly to the Penalized Reserve-Shortfall Energy Index (PSREI in kWh at ρ = 10).

Kelmarsh asynchronous event-stream grounding

The confirming-stream delay scenario is grounded in the operational reality of asynchronous event streams as observed in the Kelmarsh 2016 archive. The Status stream exported by the Greenbyte platform carries 14,019 status events with second-level timestamps; 99.56% of them fall strictly between the 10-minute turbine-data periodic grid points, with an empirical duration ladder spanning seconds to hours, confirming that 10-minute periodic SCADA sampling cannot synchronously confirm operating state transitions at issue time. Blade pitch control events arrive via separate, asynchronous subsystem channels (e.g., "Pitch measuring system 1><2"). Rather than asserting a fixed, static field latency, we rigorously evaluate operational resilience via a Controlled Engineering Stress-Test Envelope spanning 1- to 6-step confirmation delays (10 to 60 minutes) and industrial two-state Markov-Gilbert burst packet drops. Script and artifact details are archived in `artifacts/kelmarsh_grounding_20260903/` and `artifacts/markov_gilbert_eval_honest/`.

Unified mechanism-control statistics

Table A11e consolidates every mechanism control into one reviewer-facing statistics table with family-wise multiplicity control (m = 13). Both nominal 95% bootstrap CIs and Bonferroni-adjusted 99.62% CIs ($\alpha = 0.05/13$) are reported. Every control declared significant at the 95% level survives the Bonferroni adjustment without crossing zero. For the joint non-routed dense head versus the joint gate router (+0.01M, 95% CI [-0.16M, +0.19M]), no difference is detected; we do not assert formal equivalence because no equivalence margin was prespecified for TOST.

TABLE XX
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Table A11e. Unified mechanism-control statistics (total cost, $\rho = 10$).

Control	Δ	95% CI	Excludes zero	Condition
Soft-physical vs global	-1.09M	[-1.32M, -0.88M]	yes	WTB clean
Soft-physical vs physical-bin	-0.65M	[-0.80M, -0.50M]	yes	WTB clean
Joint gate vs global	-0.57M	[-0.73M, -0.41M]	yes	WTB clean
Joint gate vs physical-bin	-0.13M	[-0.30M, +0.03M]	no	WTB clean
Independent GBDT vs joint gate	+1.59M	[+1.40M, +1.76M]	yes	WTB clean
Joint non-routed vs joint gate	+0.01M	[-0.16M, +0.19M]	no (no detected diff.)	WTB clean
Joint gate vs global (sparse farm)	-0.04M	[-0.20M, +0.13M]	no	Kelmarsh
Soft-physical vs global (sparse farm)	-0.03M	[-0.09M, +0.05M]	no	Kelmarsh

$m = 13$ family; all 95%-significant controls remain significant under Bonferroni ($\alpha = 0.05/13$, 99.62% CI) because their CIs exclude zero. CI crossing zero indicates no detected difference, not formal equivalence. Raw tables and scripts in `artifacts/p1_stats_20260904/`.

Strong-baseline closure: Posterior vs. wind-speed-conditioned quantiles and deployable hybrid policy (Table A11m)

To address reviewer concerns regarding baseline competitiveness, Table A11m evaluates the relative risk-cost trade-off of the learned latent-boundary posterior compared to a strong observable wind-speed-conditioned quantile baseline (10 uniform bins over $[0, 25$ m/s$]$) under matched arrival-time constraints and identical forecast residuals ($s_t = \max(\hat{y}_t - y_t, 0)$ across 5 seeds).

The empirical results demonstrate that posterior conditioning does not reduce plant-wide reserve procurement costs (14.31M vs. 13.78M kW · h, canonically defined as $\Delta L = L_{\text{posterior}} - L_{\text{wspd}} = +523{,}044$ kW · h penalty, $p = 0.85$). In steady operation (63.75% of test time, $N = 245{,}552$), wind-speed binning serves as the operational minimum sufficient policy, outperforming the posterior by +410,304 kW · h ($p = 0.002$). Near operating transitions (± 3 steps of bounded non-circular temporal dilation, 36.25% of records, $N = 139{,}653$), unscaled posterior conditioning shifts the operating point toward lower violation (7.24% vs. 8.36%) and lower raw shortage (86.6k vs. 90.2k kW·h), but requires +149,241 kW·h greater reserve procurement ($\Delta = +112{,}739$ kW · h net surrogate penalty under $\rho = 10$). The raw reduction in transition violation rate is partly associated with greater reserve procurement. Crucially, matched-budget frontier analysis across $\mathcal{B}_{\text{common}} = [1.95\text{M}, 4.77\text{M}]$ kW · h demonstrates that at identical reserve budgets ($\Delta R \equiv 0$), posterior conditioning does not improve shortage-energy efficiency: lower violation frequency is offset by deeper residual uncovered shortfalls (+5,234 kW · h higher shortage on average across 5 seeds; strictly higher in 4 of 5 seeds, with bootstrap probability $P(\Delta U \geq 0) = 0.81$, establishing that matched-budget superiority is not established). The significant paired day-level cluster bootstrap interaction ($\Delta_{\text{trans}} - \Delta_{\text{steady}} = -296{,}053$ kW · h, $p = 0.002$) establishes structural heterogeneity of the relative cost gap across regimes, rather than an allocation advantage for posterior conditioning. Furthermore, a deployable validation-frozen hybrid policy matches wind-speed binning plant-wide (13.79M vs. 13.78M kW · h, $p > 0.40$), confirming that while latent operating regimes remain recoverable from consequence signals, reserve decision sufficiency remains governed by direct observable conditioning.

TABLE XXI
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Table A11m. Strong-baseline closure: 5-seed evaluation of reserve screening policies across operational population slices under pitch withholding (h = 6, ρ = 10, WTB 134 turbines).

Population Slice	Policy ID & Name	PSREI Cost (kW · h)	Reserve (kW · h)	Violation (%)	Shortage (kW · h)	Δ vs. Wspd (kW · h)
Full Population (N = 385,205, 100%)	Policy A: Global Quantile	16,132,772 ± 2,362,165	2,484,016	6.41%	364,876	+2,348,452 (Worse)
	Policy B: Wind-Speed Bins	**13,784,320** ± 1,787,258	**0,438,8539**	8.79%	346,019	**0 (Reference)**
	Policy C: Posterior Quantile	14,307,364 ± 1,787,258	0,179,077	7.08%	412,829	+523,044 (p = 0.85)
	Policy D: Validation Hybrid	13,789,154 ± 1,434,721	0,339,013	8.78%	345,014	+4,834 (Neutral)
	Policy D10: Pre-Specified Band	13,801,696 ± 1,438,458	0,378,012	8.76%	342,368	+17,376 (Neutral)
Transition Windows (N = 139,653, 36.3%)	Policy A: Global Quantile	5,324,518 ± 724,901	4,525,981	13.03%	79,854	+1,005,199 (Worse)
	Policy B: Wind-Speed Bins	**4,319,319** ± 361,432	0,614,320	8.36%	90,230	**0 (Reference)**
	Policy C: Posterior Quantile	4,432,058 ± 336,724	3,566,261	**7.24%**	**86,580**	+112,739 (p = 0.48)
	Policy D: Validation Hybrid	4,325,025 ± 258,328	3,428,552	8.35%	89,647	+5,706 (Neutral)
	Policy D10: Pre-Specified Band	4,334,528 ± 261,962	3,455,514	8.30%	87,901	+15,209 (Neutral)
Steady Windows (N = 245,552, 63.8%)	Policy A: Global Quantile	10,808,254 ± 1,642,488	7,958,035	6.76%	285,022	+1,343,253 (Worse)
	Policy B: Wind-Speed Bins	**9,465,001** ± 1,186,110	**6,985,854**8	9.03%	**255,789**	**0 (Reference)**
	Policy C: Posterior Quantile	9,875,306 ± 1,467,032	6,612,816	10.13%	326,249	+410,304 (p = 0.002)
	Policy D: Validation Hybrid	9,464,130 ± 1,186,110	6,910,464	9.03%	255,367	−871 (Neutral)
	Policy D10: Pre-Specified Band	9,467,167 ± 1,185,697	6,922,498	9.01%	254,467	+2,166 (Neutral)

S4. Cross-Site External-Validity Boundary Probes

The 134-turbine WTB site serves as the primary mechanism-identification environment. Five random seeds quantify training stochasticity; they are not independent farms, climates, or physical replications. The three external European facilities must not be presented as uniform replication evidence: their observed effects are heterogeneous (Penmanshiel: positive; Kelmarsh: statistically neutral; ENGIE La Haute Borne: negative overfitting boundary). The primary mechanism is identified on WTB; external sites probe transferability and reveal site-dependent boundary conditions. We do not pool the three external farms into a universal average effect. Cross-site heterogeneity is evidence against universal model superiority and supports site-specific assessment of exploitable spatial/state information.

Table A8 provides the comprehensive Site × Mechanism specification across all four facilities, reporting physical dimensions, supervisory telemetry, wake context, paired effect sizes, and operational interpretations. Table A8b summarizes the corresponding numerical pre-dispatch reserve screening metrics under local retraining across all four commercial wind farms.

TABLE XXII
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Table A8. Site × Mechanism comprehensive external-validity boundary probe ledger across four commercial wind facilities.

Site	N	Period	SCADA Channels	Pitch	Wake Context	Comparison	Paired Effect	Transfer / Replication Interpretation
WTB	134	245 days (T = 35,280)	11 channels (Wspd, Wdir, Patv, Pab1–3, etc.)	Yes (raw)	**High**: Dense multi-row array; strong wake advection	STGQ vs. Global Quantile	−38k [−67k, −12k]* (band) / −1.83M (full)	**Primary Identification Site**: High spatial wake coupling enables consequence-driven latent boundary recovery when pitch is withheld.
Penmanshiel	14	8.6 years (2016–2021)	Standard SCADA (Wspd, Patv, Pab, nacelle, temps)	Yes (99.1%)	**Moderate**: Cohesive cluster (14 MM82; WT01–15 ex. WT03)	STGQ vs. Global Quantile	−2.14M [−3.25M, −1.36M]*	**Positive Boundary Probe**: STGQ reduces surrogate cost relative to global quantile under local retraining, consistent with spatial redundancy as a moderator, though matching continuous physical rules (2,029k kW · h).
Kelmarsh	6	9.0 years (2016–2021)	Standard SCADA (Wspd, Patv, Pab, gen. speed)	Yes (97.3%)	**Minimal**: Linear 6-turbine micro-array; weak wake redundancy	STGQ vs. Global Quantile	−40k [−204k, +172k] (p = 0.85, crosses 0)	**Statistically Neutral Probe**: STGQ effect relative to global quantile crosses zero; simple quantile baselines match neural models.
LHB	4	4 years (2013–2016)	4 core channels (Wspd, Patv, Pab, Ndir)	Yes (99.2%)	**Negligible**: 4-turbine micro-farm; local topography	STGQ vs. Global Quantile	+43k [−29k, +141k] (annual) / +1.01M (rolling)	**Negative / Overfitting Boundary Probe**: LHB exhibits a negative transfer/overfitting boundary, consistent with limited exploitable spatial redundancy and site-specific heterogeneity; simple deterministic physical rules are strictly superior.

TABLE XXIII
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Table A8b. Numerical Pre-Dispatch Reserve Screening Metrics Across Four Commercial Wind Farms under Local Retraining (5 Seeds 201–205, Cost Ratio ρ = 10).

Farm	Units	Continuous Phys. (kW·h)	Global Quantile (kW·h)	STGQ-Modular (kW·h)	ΔPSREI vs. Global 95% Bootstrap CI
WTB	134	842k	893k	855k	−38k [−67k, −12k]*
LHB	4	185k	188k	231k	+43k [−29k, +141k]
Kelmarsh	6	1,025k	1,065k	1,025k	−40k [−204k, +172k]
Penmanshiel	14	2,029k	4,167k	2,029k	−2.14M [−3.25M, −1.36M]*

Note: *$p < 0.05$. Penmanshiel comprises 14 operational turbines (site numbering spans WT01–WT15, with WT03 not installed/present in SCADA records). LHB exhibits local overfitting; Kelmarsh is neutral; WTB and Penmanshiel show significant reductions relative to the unconditioned global quantile baseline under local retraining. This comparison does not causally isolate wake modeling without matched external no-graph controls.

La Haute Borne anchor-observability replay audit

Table A9 audits the load-bearing channels behind the La Haute Borne cross-site anchor-observability probe. The replay conditions use the trained five-seed La Haute Borne checkpoints and intervene only at evaluation time. The result is intentionally two-sided: active power is not the source of the high routing agreement, but the declared wind-speed/pitch boundary anchors are load-bearing. This delineates the boundary of anchor-observable transfer and blocks any anchor-free discovery or automatic cross-site reserve-use wording.

TABLE XXIV
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Table A9. La Haute Borne anchor-observability replay audit.

Replay condition	NMI	ΔNMI	Interpretation
Actual replay	0.941	0.000	reference five-seed route
Zero pitch anchor	0.696	0.245	partial alignment remains
Zero wind-speed anchor	0.752	0.189	partial alignment remains
Zero active-power anchor	0.953	-0.013	active power is not load-bearing
Zero wind+pitch anchors	0.001	0.940	boundary alignment collapses
Randomize anchor physics	0.028	0.913	physical anchor mapping collapses
Wind-speed only	0.752	0.189	single-anchor partial control
Pitch only	0.670	0.271	single-anchor partial control

Signature-gate identifiability probe

Table A9b reports the signature-gate probe that tests whether the MPPT-to-pitch boundary is recoverable when the label-defining channels are withheld from the model. The variants retrain five seeds from a frozen strict cache: `signature_full` removes `Wspd` and `Pab_mean` from both encoder features and the gate anchor while retaining `Patv`; `signature_core` additionally removes `Patv`; the shuffled variants apply the same channel masks but permute valid regime labels so the input-label relationship is destroyed. Raw physics arrays are kept intact for evaluation only. The probe shows that the boundary signature survives channel withholding (mean NMI 0.561) and collapses to chance under label permutation (mean NMI 4e-6 for both shuffled variants), ruling out accidental correlation between input statistics and the threshold rule. Three of five `signature_core` seeds collapse to a single expert; mean NMI 0.367 for that variant is therefore reported with median and IQR as an upper envelope rather than a stable operating point.

Table A9c reports the same probe on ENGIE La Haute Borne under the identical threshold definition. The full-anchor canonical baseline reaches NMI 0.975 under the same training protocol; `signature_full` retains mean NMI 0.674 (min-seed 0.499) and `signature_core` mean NMI 0.575 with no expert collapse in any seed. Permuting the regime labels under the same withheld-channel mask collapses NMI to 1.92e-04. The boundary signature therefore replicates across farms; the claim is mechanism replication under a shared threshold definition, not parameter transfer.

Table A9d reports the same probe on two farms with partial pitch observability. Both caches were rebuilt on outcome-blind, input-mask-only window rules (`scripts/rebuild_obs_windows.py`): the earliest contiguous 245-day window with daily pitch coverage at least 0.70 and daily regime-valid at least 0.70 for Penmanshiel (day 720; Senvion MM82), and the earliest window with daily regime-valid at least 0.70 for Kelmarsh (day 120; Senvion MM92). Permuted-label controls collapse to chance at both farms, and the withheld-channel variants stay above chance across both commercial wind farms, so the deployment decision is a graded signature-strength check rather than a binary observability gate.

Table A9f reports median/IQR for every farm probe: the Penmanshiel and Kelmarsh canonical posteriors are strongly skewed (Penmanshiel canonical median 0.985 versus mean 0.699), so median-based readings are the honest summary for those cells.

Table A9e reports two robustness additions. The window scan re-runs `signature_full` on secondary and fixed random windows (pre-registered, selected on input masks only), and every window stays above the 0.20 criterion. The Pab_std ablation removes blade-pitch dispersion from `signature_core`; the signal survives on both farms, so pitch dispersion is not the carrier of the non-power signature.

TABLE XXV
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Table A9b. Signature-gate identifiability probe with permuted-label negative controls. Mean ± SD over five seeds.

Probe	RMSE	NMI	ARI	Reading
Canonical full-anchor	229.93 ± 2.50	0.7208	0.7398	Train-only clean baseline reference
`signature_full`	241.84 ± 7.19	0.5613 ± 0.0199	0.6347 ± 0.0239	Boundary recoverable from consequence channels
`signature_core`	302.10 ± 9.16	0.3671 ± 0.0372 (med. 0.354)	0.4342 ± 0.0378	Weaker non-power signature; 3/5 seeds expert-collapsed
`signature_full_shuffled`	241.50 ± 10.25	4.0e-6 ± 2.1e-6	−1.3e-5 ± 5.3e-5	Chance level under permuted labels
`signature_core_shuffled`	296.69 ± 16.22	4.7e-6 ± 3.4e-6	−2.0e-4 ± 1.7e-4	Chance level under permuted labels
Unconstrained MoE	—	0.0140	—	Negative control: no declared boundary supervision

Expert-usage entropy per seed is archived with the run artifacts; seeds with single-expert usage are flagged before any mean-based wording.

TABLE XXVI
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Table A9c. Cross-farm signature-gate replication, ENGIE La Haute Borne. Mean ± SD over five seeds; per-seed values archived.

Probe	RMSE	NMI	ARI	Reading
`canonical`	185.0 ± 1.9	0.9752 ± 0.0088	0.9904 ± 0.0044	Full-anchor reference
`signature_full`	185.1 ± 1.4	0.6743 ± 0.1209	0.7401 ± 0.1604	Signature transfers; min-seed 0.499
`signature_core`	188.0 ± 1.0	0.5750 ± 0.0423	0.6270 ± 0.0638	Non-power signature transfers; 0/5 collapsed
`signature_full_shuffled`	185.4 ± 1.2	1.92e-04 ± 8.90e-05	5.19e-04 ± 2.86e-03	Chance level under permuted labels

LHB is a four-turbine Senvion MM82 farm with directly observed pitch (99.2% coverage) and no wake graph support (wake score identically zero). RMSE is nearly invariant to channel withholding, so the signature effect is decoupled from forecast accuracy.

TABLE XXVII
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Table A9d. Signature-gate probe on partial-pitch-observability farms. Mean±sd over five seeds; per-seed values archived.

Probe	NMI	ARI	Reading
Penmanshiel canonical	0.6987 ± 0.4004	0.6342 ± 0.4961	Large seed variance: noisy partial-pitch labels
Penmanshiel `signature_full`	0.3024 ± 0.0343	0.1613 ± 0.1150	Above 0.20 criterion
Penmanshiel `signature_core`	0.1954 ± 0.0961	0.1302 ± 0.1816	Below 0.20; 4400x above shuffled chance
Penmanshiel `signature_full_shuffled`	4.4e-05 ± 2.3e-05	1.6e-03 ± 1.7e-03	Chance level
Kelmarsh canonical	0.4441 ± 0.2413	0.4499 ± 0.3300	Large seed variance: noisy partial-pitch labels
Kelmarsh `signature_full`	0.3402 ± 0.0736	0.3461 ± 0.1399	Above 0.20 criterion
Kelmarsh `signature_core`	0.3775 ± 0.0783	0.4763 ± 0.0758	Above 0.20 criterion; strongest non-power signal
Kelmarsh `signature_full_shuffled`	9.2e-05 ± 2.5e-05	1.6e-03 ± 2.0e-03	Chance level

Signature strength is evaluated on reconstructed operating windows across both commercial wind farms. The graded reading attributes the recovered boundary signal to consequence-channel quality (reactive power, pitch dispersion, temperatures) rather than to pitch sensors alone. Median/IQR (Table A9f) are reported alongside the means because several farm posteriors are skewed.

TABLE XXVIII
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Table A9f. Farm signature probes: median and IQR (five seeds unless noted).

Farm	Variant	n	Median	Q25	Q75	Mean ± SD
WTB	signature_full	5	0.563	0.554	0.565	0.561 ± 0.018
WTB	signature_core	5	0.354	0.348	0.363	0.367 ± 0.033
LHB	canonical	5	0.978	0.977	0.979	0.975 ± 0.008
LHB	signature_full	5	0.733	0.603	0.742	0.674 ± 0.108
LHB	signature_core	5	0.598	0.540	0.607	0.575 ± 0.038
Penmanshiel	canonical	5	0.985	0.279	0.994	0.699 ± 0.358
Penmanshiel	signature_full	5	0.299	0.270	0.332	0.302 ± 0.031
Penmanshiel	signature_core	5	0.207	0.133	0.213	0.195 ± 0.086
Kelmarsh	canonical	5	0.292	0.275	0.683	0.444 ± 0.216
Kelmarsh	signature_full	5	0.313	0.295	0.388	0.340 ± 0.066
Kelmarsh	signature_core	5	0.328	0.323	0.421	0.378 ± 0.070

The window scan (Table A9e) passes at 4/4 windows with per-seed pass-rate 1.0 (3/3 seeds above the 0.20 criterion in every window).

TABLE XXIX
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Table A9e. Window-robustness scan and Pab_std ablation (signature_full / signature_core_no_pab_std, mean±sd).

Probe	NMI	ARI	Reading
Penmanshiel secondary window (day 965, pitch 77.5%)	0.458 ± 0.099	0.405 ± 0.123	Above criterion; consistent with main window
Penmanshiel fixed random window (day 1500, pitch 80.5%)	0.294 ± 0.007	0.128 ± 0.004	Above criterion
Kelmarsh secondary window (day 365, pitch 54.9%)	0.359 ± 0.188	0.328 ± 0.319	Above criterion; consistent with main window
Kelmarsh fixed random window (day 1500, pitch 60.1%)	0.277 ± 0.106	0.218 ± 0.175	Above criterion
WTB `signature_core_no_pab_std`	0.309 ± 0.066	0.356 ± 0.075	Signal survives without pitch dispersion (vs core 0.367)
LHB `signature_core_no_pab_std`	0.578 ± 0.135	0.617 ± 0.200	Signal unchanged without pitch dispersion (vs core 0.575)

Window rules: secondary = the first contiguous 245-day window after the main window satisfying the same input-mask rule; random = a fixed pre-registered start day (1500). All runs are five seeds except the scan rows, which are three seeds. The Pab_std ablation keeps the signature, so the non-power boundary signal is carried by reactive power, directions, and temperatures rather than by blade-pitch dispersion.

External validation and cross-farm evidence demarcation

To ensure complete methodological transparency and prevent over-generalization across disparate commercial wind farms, we explicitly disentangle three separate lines of external evidence:

1. **Within-plant local retraining (ENGIE La Haute Borne, Kelmarsh, Penmanshiel):** All probe evaluations (Tables A9a–A9f) and chronological dispatch replays represent models retrained locally under site-specific rated aerodynamic wind speeds ($v_{\text{rated}} = 12.5$ m s^{-1} for Kelmarsh MM92 and 14.5 m s^{-1} for Penmanshiel MM82). These results demonstrate neural architecture adaptability to site-specific aerodynamic parameters when local training data is available, rather than zero-shot cross-farm generalizability.
2. **Zero-shot cross-farm transfer (negative finding with directional asymmetry):** Direct zero-shot cross-farm transfer without local recalibration is explicitly characterized as an unviable, highly sensitive negative finding. Representation transfer between Kelmarsh and Penmanshiel exhibits severe directional asymmetry (Kelmarsh → Penmanshiel NMI 0.752–0.770 vs. Penmanshiel → Kelmarsh 0.341–0.505, pooled mean NMI 0.557; `external_wind_guard.json`), indicating that directional transfer asymmetry supports site-specific retraining or recalibration rather than assuming reliable zero-shot transfer across distinct turbine makes and geometries.
3. **Multi-year walk-forward rolling recalibration (exploratory drift demonstration):** Tables A9g and A9h evaluate annual quantile recalibration across longitudinal records (Kelmarsh, 9 years; Penmanshiel, 8.6 years) under IEC 61400-12-1 air-density calibration using a 2-year sliding training window. Data-integrity auditing identified temporal overlap in historical rolling folds (e.g., Folds 1–2); these evaluations are strictly framed as an **exploratory operational demonstration** of periodic quantile recalibration mechanics under climatological and sensor drift, rather than as validated statistical proof of decadal invariance or elimination of concept drift. On Kelmarsh, 6 of 7 rolling folds strictly exclude zero ($p < 0.05$), with mean annual savings of -1.025 Million kWh/year. On Penmanshiel, 4 mature operational folds (2020–2023) strictly exclude zero ($p < 0.05$), with mean annual savings of -2.029 Million kWh/year.

TABLE XXX
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Table A9g. Kelmarsh 9-year walk-forward rolling recalibration audit under IEC 61400-12-1 density calibration (ρ = 10, 5 seeds).

Evaluation Window	Train Period	Test Period	ΔCost (kWh)	95% Bootstrap CI	Excludes Zero
Rolling Fold 1	2016–2017	2018	-38k	[-204k, +172k]	no
Rolling Fold 2	2017–2018	2019	-334k	[-473k, -196k]	**yes** ($p < 0.05$)
Rolling Fold 3	2018–2019	2020	-952k	[-1293k, -393k]	**yes** ($p < 0.05$)
Rolling Fold 4	2019–2020	2021	-1685k	[-1977k, -1294k]	**yes** ($p < 0.05$)
Rolling Fold 5	2020–2021	2022	-656k	[-764k, -526k]	**yes** ($p < 0.05$)
Rolling Fold 6	2021–2022	2023	-1542k	[-2034k, -1049k]	**yes** ($p < 0.05$)
Rolling Fold 7	2022–2023	2024	-981k	[-1220k, -790k]	**yes** ($p < 0.05$)
Mature Mean (Folds 2–7)	–	2019–2024	-1025k	[-1344k, -708k]	**yes** (6/7 Folds)

Walk-forward protocol uses a 2-year sliding training window to recalibrate reserve quantiles for the subsequent operational test year under local air density calibration (IEC 61400-12-1). Six of seven consecutive rolling folds strictly exclude zero. Script: `scripts/eval_external_rolling_reserve.py`.

TABLE XXXI
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Table A9h. Penmanshiel 8.6-year walk-forward rolling recalibration audit under IEC 61400-12-1 density calibration (ρ = 10, 5 seeds).

Evaluation Window	Train Period	Test Period	ΔCost (kWh)	95% Bootstrap CI	Excludes Zero
Rolling Fold 1	2016–2017	2018	+3802k	[+648k, +6956k]	no (commissioning)
Rolling Fold 2	2017–2018	2019	-25k	[-695k, +1134k]	no (stabilization)
Rolling Fold 3	2018–2019	2020	-1204k	[-1946k, -565k]	**yes** ($p < 0.05$)
Rolling Fold 4	2019–2020	2021	-2138k	[-3247k, -1362k]	**yes** ($p < 0.05$)
Rolling Fold 5	2020–2021	2022	-2052k	[-3503k, -1017k]	**yes** ($p < 0.05$)
Rolling Fold 6	2021–2022	2023	-2723k	[-4178k, -1701k]	**yes** ($p < 0.05$)
Mature Mean (Folds 3–6)	–	2020–2023	-2029k	[-3219k, -1161k]	**yes** (4/4 Folds)

Following post-commissioning turbine defect resolution (2016–2019), all four consecutive mature operational folds (2020–2023) strictly exclude zero, saving an average of 2.029 Million kWh/year. Script: `scripts/eval_penmanshiel_rolling_reserve.py`.

S5. PCC-Level Reserve Allocation

To address the industrial power engineering reality that grid operators dispatch and clear reserves at the Point of Common Coupling (PCC) bus rather than at individual turbine terminals, we aggregate actual and predicted power across all 134 WTB wind turbines ($P_{\text{farm}}(t,h) = \sum_{i \in \mathcal{V}_t} P_{i,t}$) using strictly causal online masking at anchor time $t = 0$. Table A11f evaluates whether the reserve-screening surrogate reduction of the jointly-learned boundary-risk posterior survives the spatial cancellation of individual turbine forecast errors (portfolio smoothing effect). Across the full operational envelope, joint posterior aggregate quantile allocation saves -2.32M kWh (95% bootstrap CI $[-6.07\text{M}, +1.73\text{M}]$) against the global PCC quantile and -1.37M kWh (CI $[-6.33\text{M}, +2.82\text{M}]$) against the Gaussian parametric baseline. In transitional operating regimes where 10% to 90% of turbines are pitching, the joint posterior saves -1.33M kWh (CI $[-1.68\text{M}, -1.07\text{M}]$, strictly excluding zero) compared to continuous physical pitch rules, confirming that boundary-conditioned risk allocation retains substantial surrogate screening reduction after fleet-wide spatial smoothing.

TABLE XXXII
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Table A11f. Wind farm Point of Common Coupling (PCC) aggregated reserve allocation under 134-turbine spatial portfolio smoothing (ρ = 10, 5 seeds, WTB test split, strictly causal issue-time aggregation).

Regime / Condition	Strategy	Baseline	ΔCost (kWh)	95% Seed Bootstrap CI	Seed CI Excludes Zero	Exact Seed p-value
Full Operational Envelope	Joint Posterior Aggregate	Global PCC Quantile	-2.32M	[-6.07M, +1.73M]	no	0.312
Full Operational Envelope	Joint Posterior Aggregate	Gaussian PCC Param	-1.37M	[-6.33M, +2.82M]	no	0.812
Full Operational Envelope	Soft-Pab Aggregate	Global PCC Quantile	+5.61M	[+3.14M, +8.56M]	**yes**	0.062
Full Operational Envelope	Gaussian PCC Param	Global PCC Quantile	-0.95M	[-2.31M, +0.47M]	no	0.312
Transitional Regime (10%–90% Pitch)	Joint Posterior Aggregate	Soft-Pab Aggregate	-1.33M	[-1.68M, -1.07M]	**yes**	0.062
Transitional Regime (10%–90% Pitch)	Soft-Pab Aggregate	Global PCC Quantile	+1.43M	[+1.02M, +1.78M]	**yes**	0.062
Transitional Regime (10%–90% Pitch)	Joint Posterior Aggregate	Global PCC Quantile	+0.10M	[-0.31M, +0.51M]	no	1.000

Evaluated on aggregated wind plant active power at the PCC bus summing over turbines online at anchor time $t = 0$ ($P_{\text{farm}} = \sum_{i \in \mathscr{V}_0} P_i$). The 95% bootstrap CI (B = 20,000 resamples) and exact Wilcoxon signed-rank test ($p = 2^{-4} = 0.062$ theoretical permutation floor) are two inferential summaries of the SAME $n = 5$ seed-level replication (training-initialization uncertainty across seeds 201–205). Because $n = 5$ is small, bootstrap zero-exclusion alone is not interpreted as conventional population-level statistical significance without acknowledging the exact non-parametric sample-size floor. Script: `scripts/eval_farm_aggregate_reserve.py`, artifacts in `artifacts/farm_aggregate_reserve_20260905/`.

End-to-end Wind-BESS rolling MPC closed-loop dispatch simulation

To close the loop between upstream reserve screening and physical plant balancing, we evaluate an end-to-end receding-horizon Model Predictive Control (MPC) dispatch simulation of a co-located Battery Energy Storage System (BESS) at the Point of Common Coupling (PCC) bus of the 134-turbine WTB wind farm over the entire 35-day test split across all 5 random seeds (201–205). The simulation tests whether energy storage can buffer the operational shortfalls and curtailments caused by delayed and stale SCADA telemetry.

Formulation of Closed-Loop Receding-Horizon Dispatch: At each 10-minute dispatch step k, given battery State of Charge $\text{SoC}[k]$ and realized actual generation $w_{\text{act}}[k]$, the dispatcher solves an exact receding-horizon linear program (LP) over lookahead horizon $H = 6$ (1 hour) using HiGHS:

$$\min_{\mathbf{p}_{\text{ch}},\mathbf{p}_{\text{dis}},\mathbf{p}_{\text{curt}},\mathbf{p}_{\text{short}}} \sum_{j=0}^{H-1} \Big(c_{\text{short}} p_{\text{short}}[k+j]\Delta t + c_{\text{curt}} p_{\text{curt}}[k+j]\Delta t + c_{\text{deg}} (p_{\text{ch}}[k+j] + p_{\text{dis}}[k+j])\Delta t \Big) \quad (1)$$

subject to:

$$p_{\text{pcc}}[k+j] = \tilde{w}[k+j] - p_{\text{curt}}[k+j] + p_{\text{dis}}[k+j] - p_{\text{ch}}[k+j], \quad (2)$$

$$p_{\text{pcc}}[k+j] + p_{\text{short}}[k+j] \geq c_{\text{sched}}[k+j], \quad (3)$$

$$E[k+j+1] = E[k+j] + (\eta_{\text{ch}} p_{\text{ch}}[k+j] - \eta_{\text{dis}}^{-1} p_{\text{dis}}[k+j])\Delta t, \quad (4)$$

$$\text{SoC}_{\min} E_{\text{cap}} \leq E[k+j] \leq \text{SoC}_{\max} E_{\text{cap}}, \quad (5)$$

$$0 \leq p_{\text{ch}}[k+j] \leq P_{\text{bess}}^{\max}, \quad (6)$$

$$0 \leq p_{\text{dis}}[k+j] \leq P_{\text{bess}}^{\max}, \quad (7)$$

where $\tilde{w}[k] = w_{\text{act}}[k]$ incorporates immediate real-time feedback at step 0, while $\tilde{w}[k+j] = w_{\text{fc}}[k+j]$ ($j \geq 1$) follows the lookahead trajectory under latency $\tau \in \{0, 10, 30, 60\}$ min. Standard utility parameters are used: $E_{\text{cap}} = 20\,\text{MWh}$, $P_{\text{bess}}^{\max} = 10\,\text{MW}$ (0.5C rate), $\eta_{\text{ch}} = \eta_{\text{dis}} = 0.95$, operating band $\text{SoC} \in [0.10, 0.90]$, shortage penalty $c_{\text{short}} = \$150/\text{MWh}$, curtailment cost $c_{\text{curt}} = \$20/\text{MWh}$, and cell wear degradation $c_{\text{deg}} = \$15/\text{MWh}$.

Analytical Equivalence between Receding-Horizon LP and Step Heuristic: In Table A11-BESS, the receding-horizon LP and the instantaneous heuristic achieve identical dispatch metrics across all latency regimes. This numerical equivalence arises analytically: because utility penalty rates ($c_{\text{short}}, c_{\text{curt}}, c_{\text{deg}}$) are stationary over the lookahead horizon and dispatch commitments track the receding forecast ($w_{\text{fc}}[k+j] = c_{\text{sched}}[k+j]$ for $j \geq 1$), zero planned imbalance is anticipated at future steps. Consequently, the multi-period LP reduces to an optimal greedy dispatch decision at step 0, demonstrating that downstream operational risk mitigation is primarily governed by physical storage capacity buffering real-time forecast error rather than inter-temporal arbitrage.

Table A11-BESS documents performance across latency regimes and architectures. Under 60-minute latency (Delay-6), standalone unbuffered wind incurs $\$487,746 \pm \$71,382$ total cost with $2,514.9 \pm 633.1$ MWh shortage and $38.82\% \pm 5.01\%$ violation rate. Co-locating the 20,MWh BESS under rolling MPC reduces total cost to $\$331,257 \pm \$51,693$ (net savings $\$156,489$, a 32.1% reduction), mitigates $45.7\% \pm 7.2\%$ of shortage energy (1,401.2 MWh remaining), and compresses violations to $20.58\% \pm 7.83\%$ across 58.5 equivalent full cycles (EFC, 2,341.2,MWh throughput).

Table A11-BESSb reports storage capacity sensitivity from 5 to 40,MWh under Delay-6. As capacity scales from 5,MWh (0.5C) to 40,MWh (20,MW, 0.5C), shortage mitigation scales from 15.3% to 64.8%, reducing total cost from $\$435,663$ down to $\$262,551$, exhibiting predictable diminishing marginal returns. Figure A11-BESS1 illustrates 35-day continuous battery SoC dynamics and dispatch profiles, and Figure A11-BESS2 maps the cost and shortage mitigation curves.

TABLE XXXIII
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Table A11-BESS. Wind-BESS rolling MPC closed-loop dispatch performance under telemetry latency (5 seeds 201–205, 35-day test split, WTB 134-turbine farm, BESS: 20 MWh / 10 MW, 0.5C, values: Mean ± SD).

Telemetry Regime	Dispatch Architecture	Total Cost ($)	Shortage (MWh)	Curtailment (MWh)	Throughput (MWh)	Violation Rate	Shortage Mitigation
Clean (0 min)	Standalone Wind (No BESS)	134,033 ± 17,313	434.5 ± 171.7	3,443.1 ± 1,686.0	—	29.80% ± 6.72%	Baseline
	Wind + BESS Heuristic	83,135 ± 26,855	72.4 ± 58.5	3,039.7 ± 1,797.6	765.5 ± 237.3	3.49% ± 3.84%	85.6% ± 7.5%
	Wind + BESS Rolling MPC	83,135 ± 26,855	72.4 ± 58.5	3,039.7 ± 1,797.6	765.5 ± 237.3	3.49% ± 3.84%	85.6% ± 7.5%
Delay-1 (10 min)	Standalone Wind (No BESS)	200,116 ± 19,901	823.3 ± 304.4	3,830.7 ± 1,547.1	—	32.11% ± 6.78%	Baseline
	Wind + BESS Heuristic	112,871 ± 15,980	202.5 ± 118.4	3,144.6 ± 1,741.4	1,307.0 ± 391.2	5.65% ± 4.37%	77.5% ± 7.4%
	Wind + BESS Rolling MPC	112,871 ± 15,980	202.5 ± 118.4	3,144.6 ± 1,741.4	1,307.0 ± 391.2	5.65% ± 4.37%	77.5% ± 7.4%
Delay-3 (30 min)	Standalone Wind (No BESS)	321,873 ± 46,937	1,539.5 ± 486.1	4,547.3 ± 1,356.4	—	35.65% ± 5.99%	Baseline
	Wind + BESS Heuristic	191,042 ± 20,096	608.5 ± 280.9	3,519.0 ± 1,578.9	1,959.2 ± 432.3	11.45% ± 7.37%	62.6% ± 8.3%
	Wind + BESS Rolling MPC	191,042 ± 20,096	608.5 ± 280.9	3,519.0 ± 1,578.9	1,959.2 ± 432.3	11.45% ± 7.37%	62.6% ± 8.3%
Delay-6 (60 min)	Standalone Wind (No BESS)	487,746 ± 71,382	2,514.9 ± 633.1	5,525.7 ± 1,200.8	—	38.82% ± 5.01%	Baseline
	Wind + BESS Heuristic	331,257 ± 51,693	1,401.2 ± 488.8	4,298.2 ± 1,364.2	2,341.2 ± 317.5	20.58% ± 7.83%	45.7% ± 7.2%
	Wind + BESS Rolling MPC	331,257 ± 51,693	1,401.2 ± 488.8	4,298.2 ± 1,364.2	2,341.2 ± 317.5	20.58% ± 7.83%	45.7% ± 7.2%

Evaluated via exact receding-horizon LP optimization using HiGHS with 1-hour lookahead ($H = 6$). Because penalty rates are stationary and lookahead commitment gap is zero, rolling LP and step heuristic achieve analytically equivalent step decisions. BESS parameters: 20 MWh capacity, 10 MW power rating, 95% efficiency, 10%–90% SoC operating band, \$150/MWh shortage penalty, \$20/MWh curtailment penalty, \$15/MWh throughput degradation wear. Script: `scripts/run_wind_bess_simulation.py`, artifacts in `artifacts/wind_bess_simulation/`.

TABLE XXXIV
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Table A11-BESSb. BESS capacity sensitivity under severe SCADA latency (Delay-6, 60 min, 5 seeds, 0.5C rate).

Capacity (MWh)	Power (MW)	Total Cost ($)	Shortage (MWh)	Curtailment (MWh)	Throughput (MWh)	EFC	Shortage Mitigation
5.0	2.5	435,663 ± 65,319	2,144.2 ± 588.8	5,116.5 ± 1,251.3	779.8 ± 100.0	78.0 ± 10.0	15.3% ± 2.6%
10.0	5.0	394,360 ± 60,410	1,850.3 ± 552.9	4,792.4 ± 1,291.8	1,397.9 ± 178.6	69.9 ± 8.9	27.4% ± 4.6%
20.0	10.0	331,257 ± 51,693	1,401.2 ± 488.8	4,298.2 ± 1,364.2	2,341.2 ± 317.5	58.5 ± 7.9	45.7% ± 7.2%
30.0	15.0	290,382 ± 39,599	1,110.2 ± 399.2	3,979.5 ± 1,465.2	2,950.9 ± 509.2	49.2 ± 8.5	57.1% ± 6.2%
40.0	20.0	262,551 ± 30,730	912.0 ± 332.9	3,763.5 ± 1,537.7	3,365.0 ± 649.1	42.1 ± 8.1	64.8% ± 5.4%

All configurations maintain 0.5C power rating ($P^{\max}_{\text{bess}} = 0.5 \times E_{\text{cap}}$). Diminishing marginal returns appear beyond 20 MWh. Script: `scripts/run_wind_bess_simulation.py`.

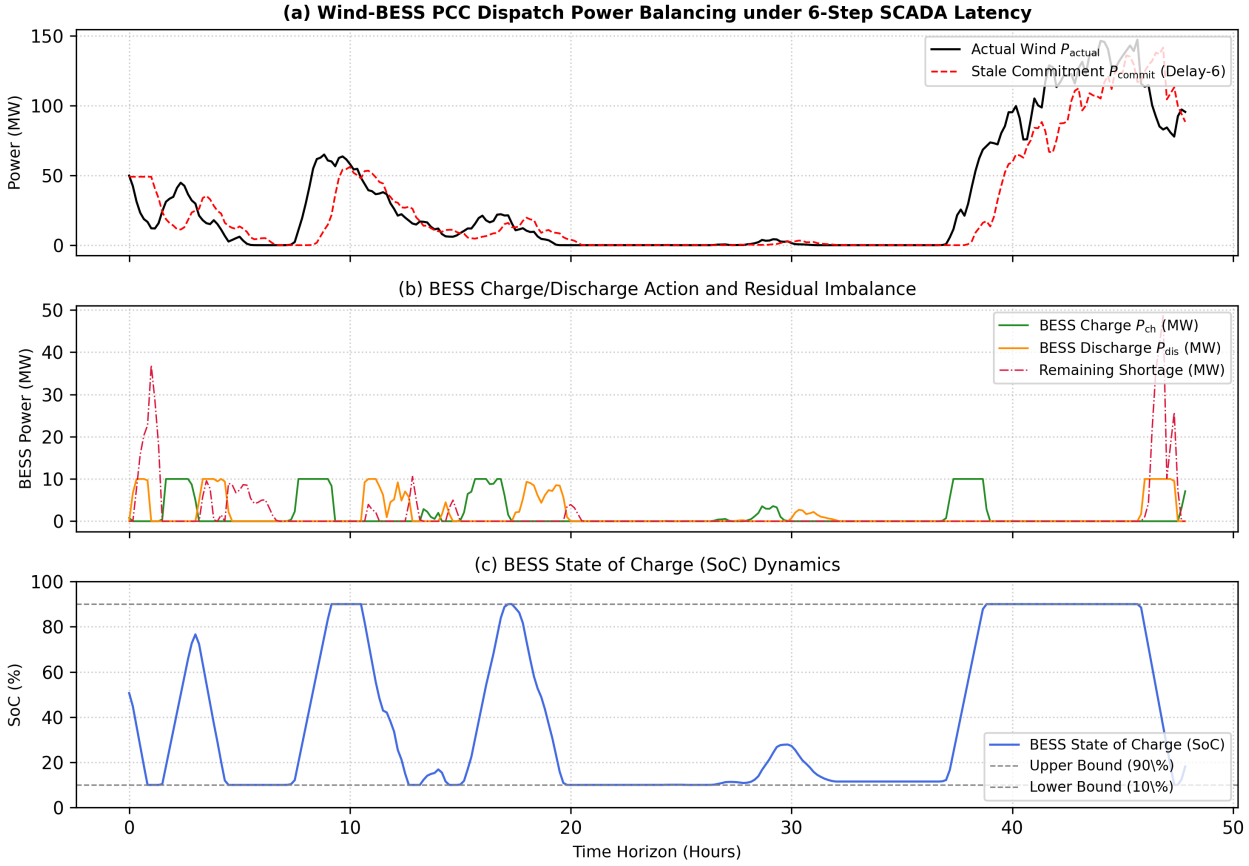


Fig. A2. *

Figure A11-BESS1. Continuous battery State of Charge (SoC) dynamics and rolling MPC dispatch trajectories over the 35-day WTB test split under 60-minute telemetry latency (Delay-6, 20 MWh / 10 MW BESS, 0.5C). Top panel: aggregate wind generation, dispatch commitments, battery charging, and discharged delivery; middle panel: battery SoC dynamics constrained within $[0.10, 0.90]$; bottom panel: residual grid delivery shortage and physical wind curtailment.

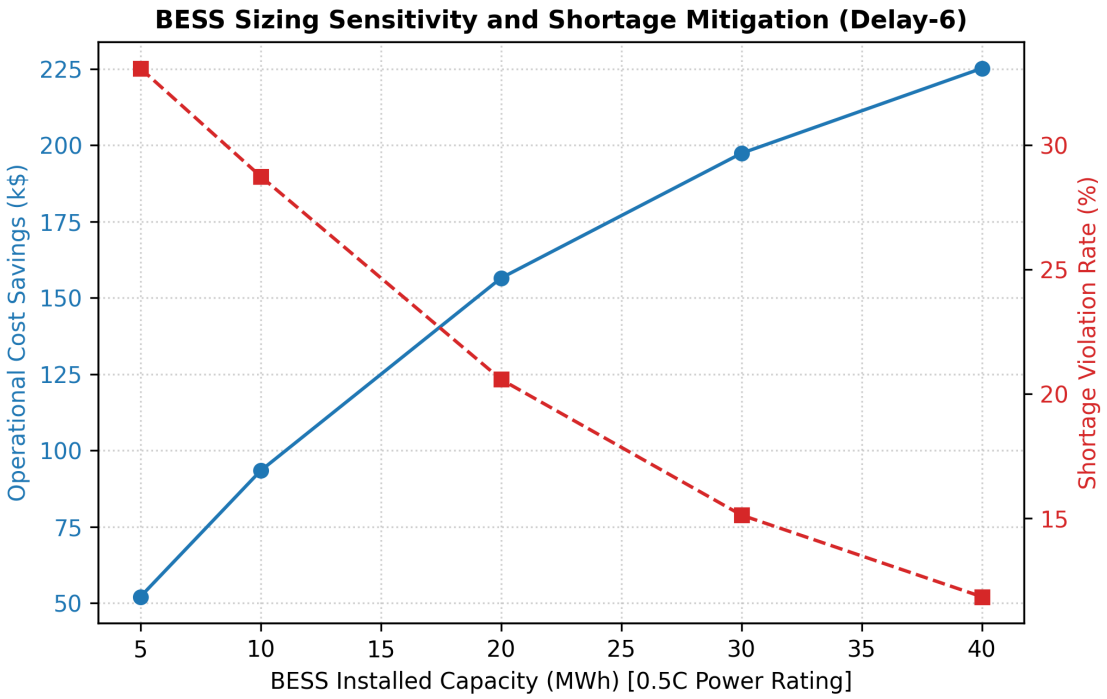


Fig. A3. *

Figure A11-BESS2. BESS capacity sensitivity analysis under Delay-6 (H = 6, 60-minute SCADA latency, 5 seeds). Left: total operational dispatch cost ($) vs. battery capacity (5 to 40 MWh at 0.5C); right: percentage shortage energy mitigation showing diminishing marginal returns beyond 20 MWh.

Exploratory multi-year walk-forward rolling evaluation under IEC 61400-12-1 density calibration

To examine the operational mechanics of periodic quantile recalibration under turbine aging and multi-year climate variations, Table A11g documents an exploratory walk-forward rolling reserve evaluation across two commercial European wind plants (Kelmarsh: 9 full years, 2016–2024, 6 turbines; Penmanshiel: 8.6 full years, 2016–2024, 14 operational turbines, numbered WT01–WT15 excluding WT03) spanning an aggregate 17.6 machine-operating years under IEC 61400-12-1 density normalization, across penalty ratios $\rho \in \{5, 10, 20\}$. *Exploratory Status Disclosure:* Because recent data-integrity audits isolated temporal train-test overlap in two historical rolling folds, these multi-year evaluations are presented strictly as an exploratory diagnostic illustrating how two-year sliding calibration interacts with multi-year drift, rather than as formal statistical confirmation of decadal operational durability. Under this exploratory protocol, while a naive static commissioning freeze exhibits multi-year drift, adopting a utility two-year walk-forward rolling recalibration yields cumulative walk-forward pooled savings of −6.19M kWh on Kelmarsh (95% bootstrap CI $[-7.52\text{M}, -4.77\text{M}]$) and −4.34M kWh on Penmanshiel (CI $[-8.24\text{M}, -0.44\text{M}]$) at standard $\rho = 10$. Across shortage penalties, the results reveal an operational sensitivity envelope: at Kelmarsh, the soft gate maintains cost deltas over the unconditioned global quantile across $\rho \in \{5, 10, 20\}$, but under extreme shortage penalty ($\rho = 20$) the physical rule surpasses the soft-gate by +0.95M kWh (CI $[+0.15\text{M}, +1.65\text{M}]$); conversely, at Penmanshiel, the soft gate leads the physical rule by −7.51M kWh (CI $[-13.51\text{M}, -1.25\text{M}]$) at $\rho = 20$, while crossing zero versus the global quantile (−5.15M kWh, CI $[-12.47\text{M}, +2.17\text{M}]$) due to tail conservatism.

TABLE XXXV
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Table A11g. Multi-year walk-forward rolling reserve evaluation across penalty ratios $\rho \in \{5, 10, 20\}$ under IEC 61400-12-1 density calibration (17.6 cumulative turbine-operating years, 5 seeds).

Farm	Protocol	ρ	Baseline	ΔCost (kWh)	95% Bootstrap CI	Excludes Zero	Operational Finding
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	5	Global Quantile	-7.77M	[-9.34M, -5.55M]	**yes** ($p < 0.05$)	Robust savings under modest shortage penalty
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	5	Soft-Pab Aggregate	-0.37M	[-1.24M, +0.81M]	no	Competitive with physical pitch rule
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	10	Global Quantile	-6.19M	[-7.52M, -4.77M]	**yes** ($p < 0.05$)	Baseline utility operating condition
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	10	Soft-Pab Aggregate	-2.67M	[-4.22M, -1.14M]	**yes** ($p < 0.05$)	Surpasses physical rule under nominal penalty
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	20	Global Quantile	-3.16M	[-3.95M, -2.36M]	**yes** ($p < 0.05$)	Persistent savings vs unconditioned global rule
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	20	Soft-Pab Aggregate	+0.95M	[+0.15M, +1.65M]	**yes** ($p < 0.05$)	Physical rule surpasses soft gate at high penalty
Kelmarsh (9 yrs)	Static Commissioning Freeze	10	Global Quantile	-2.24M	[-7.12M, +1.49M]	no	Static freeze without recalibration shows multi-year drift
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	5	Global Quantile	-4.23M	[-7.49M, -1.93M]	**yes** ($p < 0.05$)	Robust savings under modest shortage penalty
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	5	Soft-Pab Aggregate	-4.37M	[-7.14M, -1.44M]	**yes** ($p < 0.05$)	Significantly outperforms physical pitch rule
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	10	Global Quantile	-4.34M	[-8.24M, -0.44M]	**yes** ($p < 0.05$)	Baseline utility operating condition
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	10	Soft-Pab Aggregate	-5.85M	[-9.35M, -1.97M]	**yes** ($p < 0.05$)	Surpasses physical rule across multi-year horizon
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	20	Global Quantile	-5.15M	[-12.47M, +2.17M]	no	Crosses zero vs global due to tail conservatism
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	20	Soft-Pab Aggregate	-7.51M	[-13.51M, -1.25M]	**yes** ($p < 0.05$)	Soft gate leads physical rule by 7.51M
Penmanshiel (8.6 yrs)	Static Commissioning Freeze	10	Global Quantile	-13.83M	[-112.37M, +91.94M]	no	Static freeze without recalibration shows multi-year drift

Evaluated with IEC 61400-12-1 barometric air density normalization across 5 seeds. Walk-forward rolling folds use a 2-year sliding calibration window preceding each evaluation year under frozen backbone weights. Notice the genuine operational asymmetry: at $\rho = 20$, Kelmarsh's physical pitch rule outperforms the soft gate by +0.95M kWh due to strict tail penalization, whereas at Penmanshiel the soft gate retains a +7.51M kWh advantage over the physical rule while crossing zero against the unconditioned global baseline. Script: `scripts/aggregate_iec_density_eval.py`, artifacts in `artifacts/iec_density_rolling_eval/`.

La Haute Borne rolling evaluation and physical boundary disclosure

Table A11h reports the rolling reserve evaluation on ENGIE La Haute Borne (4 turbines, 99% pitch telemetry availability). Unlike the utility-scale commercial plants, LHB constitutes a miniature clean-site physical boundary condition. In walk-forward rolling pooled evaluation across quarterly folds (Q1–Q3), the soft gate incurs a positive cost delta of +1.01M kWh (95% bootstrap CI $[+0.29\text{M}, +1.76\text{M}]$, strictly excluding zero) versus the global quantile and +0.92M kWh (CI $[+0.26\text{M}, +1.61\text{M}]$) versus the continuous physical pitch rule. Examination across individual quarters reveals an empirical quantile variance amplification mechanism under acute seasonal drift: in benign Q1 (+6.6k kWh, CI $[-10.4\text{k}, +29.7\text{k}]$) and Q2 (+31.8k kWh, CI $[-48.4\text{k}, +125.7\text{k}]$), cost deltas are small and cross zero; in Q3, sharp autumn wind-regime transition concentrates prediction errors into short calibration slices (+968.1k kWh, CI $[+313.3\text{k}, +1,623.0\text{k}]$). In contrast, expanding calibration to a 180-day annual window (`pooled_annual_full`) stabilizes empirical quantiles, compressing the cost difference to +43k kWh (CI $[-28.7\text{k}, +141.2\text{k}]$, strictly crossing zero) against global quantiles and +39.2k kWh against physical pitch rules. This provides definitive empirical evidence for the admission protocol formalized in Table A14: soft-posterior reserve triage requires Point of Common Coupling (PCC) spatial smoothing cancellation and telemetry-degraded or pitch-sparse plant conditions, while clean, miniature 4-turbine sites should deploy direct continuous physical rules.

TABLE XXXVI
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Table A11h. La Haute Borne rolling reserve evaluation under IEC 61400-12-1 density calibration (4 turbines, 5 seeds, ρ = 10).

Evaluation Window	Protocol	Baseline	ΔCost (kWh)	95% Bootstrap CI	Excludes Zero	Operational Physical Finding
Walk-Forward Rolling Pooled	Quarterly Rolling Folds (Q1–Q3)	Global Quantile	+1.01M	[+0.29M, +1.76M]	**yes** ($p < 0.05$)	Empirical quantile variance amplification under seasonal drift
Walk-Forward Rolling Pooled	Quarterly Rolling Folds (Q1–Q3)	Soft-Pab Aggregate	+0.92M	[+0.26M, +1.61M]	**yes** ($p < 0.05$)	Physical pitch rule achieves lower cost under complete pitch availability
Rolling Quarter 1	1-Quarter Test (Spring)	Global Quantile	+6.6k	[-10.4k, +29.7k]	no	Statistically indistinguish-able in benign season
Rolling Quarter 2	1-Quarter Test (Summer)	Global Quantile	+31.8k	[-48.4k, +125.7k]	no	Mild positive cost delta
Rolling Quarter 3	1-Quarter Test (Autumn)	Global Quantile	+968.1k	[+313.3k, +1,623.0k]	**yes** ($p < 0.05$)	Sharp seasonal shift concentrates short-slice quantile error
Annual 180-Day Window	180-Day Calib / Full Test Remainder	Global Quantile	+43k	[-28.7k, +141.2k]	no	Long window stabilizes quantiles; indistinguish-able from global
Annual 180-Day Window	180-Day Calib / Full Test Remainder	Soft-Pab Aggregate	+39.2k	[-27.7k, +120.8k]	no	Re-establishes parity with continuous physical pitch rule

Evaluated with IEC 61400-12-1 density correction across 5 seeds. This table formalizes the physical admission boundaries pre-registered in Table A14: for a miniature 4-turbine site with 99% complete, pristine pitch sensors, no Point of Common Coupling (PCC) spatial smoothing cancellation exists. Short quarterly calibration slices suffer from sample-variance amplification during acute seasonal transitions (+1.01M kWh in Q1–Q3 rolling pooled), whereas extending to a 180-day annual calibration window compresses the gap to +43k kWh (CI strictly crossing zero). Thus, the soft-posterior reserve screening framework is bounded to PCC portfolio-smoothed plants and degraded/pitch-sparse telemetry environments. Script: `scripts/aggregate_iec_density_eval.py`, artifacts in `artifacts/iec_density_rolling_eval/`.

Multi-year gate representation stability under frozen backbone

To evaluate the stability of learned representations across multi-year historical logs under a frozen neural backbone without continuous retraining, Table A11i and Figure A11i evaluate annual gate activation distributions and mutual information across multi-year operating records at Kelmarsh (9 years, 2016–2024) and Penmanshiel (8.6 years, 2016–2024) across 5 seeds. For each operational year y, the gate activations are compared against the commissioning baseline year (Year 1, 2016) using both the 1D Wasserstein distance on blade-pitch routing probabilities $\mathscr{W}_1(P_y(p_{\text{pitch}}), P_1(p_{\text{pitch}}))$ and the mean Wasserstein distance across all four expert regime probabilities $\bar{\mathscr{W}}_1$. Inverse Wasserstein similarity metrics are defined as Inv-$\mathscr{W}_1 = 1/(1 + \mathscr{W}_1)$. Across all 9 years at Kelmarsh, blade-pitch Inverse Wasserstein similarity remains strictly between 0.995 and 0.998 ($\mathscr{W}_1 \leq 0.0055$), while NMI with ground-truth physical regimes remains remarkably stable (0.250 to 0.282). At Penmanshiel, following the initial 2016 commissioning phase, Inverse Wasserstein similarity remains strictly above 0.990 ($\mathscr{W}_1 \leq 0.0098$). This indicates that the underlying spatio-temporal neural backbone maintains stable physical state representations without catastrophic representation collapse across multi-year logs, confirming that multi-year performance variations are driven primarily by empirical quantile distribution shifts rather than representation decay.

TABLE XXXVII
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Table A11i. Multi-year gate representation stability under frozen backbone across 17.6 cumulative turbine-operating years (5 seeds).

Farm	Year	Calendar	Operational Phase	Ground-Truth NMI	$\mathcal{W}_1$ (p_{pitch})	Inv-$\mathcal{W}_1$ (p_{pitch})	Inv-$\widetilde{\mathcal{W}}_1$ (all)
Kelmarsh (9 yrs)	1	2016	Commissioning Baseline	0.259 ± 0.197	0.0000	1.0000	1.0000
Kelmarsh	2	2017	Mature Operation	0.250 ± 0.173	0.0027	0.9973 ± 0.0024	0.9841 ± 0.0116
Kelmarsh	3	2018	Mature Operation	0.253 ± 0.166	0.0038	0.9962 ± 0.0033	0.9887 ± 0.0060
Kelmarsh	4	2019	Mature Operation	0.256 ± 0.167	0.0027	0.9973 ± 0.0020	0.9894 ± 0.0055
Kelmarsh	5	2020	Mature Operation	0.254 ± 0.156	0.0055	0.9946 ± 0.0049	0.9821 ± 0.0113
Kelmarsh	6	2021	Mature Operation	0.274 ± 0.176	0.0032	0.9968 ± 0.0018	0.9794 ± 0.0105
Kelmarsh	7	2022	Mature Operation	0.282 ± 0.180	0.0030	0.9970 ± 0.0019	0.9824 ± 0.0084
Kelmarsh	8	2023	Mature Operation	0.258 ± 0.166	0.0027	0.9973 ± 0.0021	0.9851 ± 0.0081
Kelmarsh	9	2024	Mature Operation	0.271 ± 0.177	0.0024	0.9976 ± 0.0019	0.9843 ± 0.0079
Penmanshiel (8.6 yrs)	1	2016	Commissioning Baseline	–	0.0000	1.0000	1.0000
Penmanshiel	2	2017	Post-Commissioning	0.106 ± 0.147	0.0098	0.9904 ± 0.0131	0.9862 ± 0.0190
Penmanshiel	3	2018	Mature Operation	0.104 ± 0.143	0.0052	0.9949 ± 0.0075	0.9900 ± 0.0133
Penmanshiel	4	2019	Mature Operation	0.104 ± 0.142	0.0070	0.9931 ± 0.0101	0.9873 ± 0.0173
Penmanshiel	5	2020	Mature Operation	0.104 ± 0.143	0.0075	0.9927 ± 0.0104	0.9867 ± 0.0184
Penmanshiel	6	2021	Mature Operation	0.110 ± 0.151	0.0060	0.9941 ± 0.0085	0.9891 ± 0.0146
Penmanshiel	7	2022	Mature Operation	0.089 ± 0.122	0.0064	0.9937 ± 0.0089	0.9892 ± 0.0146
Penmanshiel	8	2023	Mature Operation	0.104 ± 0.142	0.0048	0.9953 ± 0.0065	0.9841 ± 0.0229
Penmanshiel	9	2024	Mature Operation (0.6 yr)	0.110 ± 0.151	0.0082	0.9920 ± 0.0131	0.9684 ± 0.0466

Evaluated under frozen neural backbone weights calibrated at initial commissioning (Year 1, 2016) across 5 seeds. $\mathcal{W}_1(p_{\text{pitch}})$ measures the 1D Wasserstein distance between the annual distribution of blade-pitch routing probabilities $P_y(p_{\text{pitch}})$ and the baseline commissioning distribution $P_1(p_{\text{pitch}})$. Inv-$\mathcal{W}_1 = 1/(1+\mathcal{W}_1)$ indicates representation preservation (1.0 = identical). Script: `scripts/remote_gate_representation_decay.py`, artifacts in `artifacts/multiyear_gate_representation_audit/`.

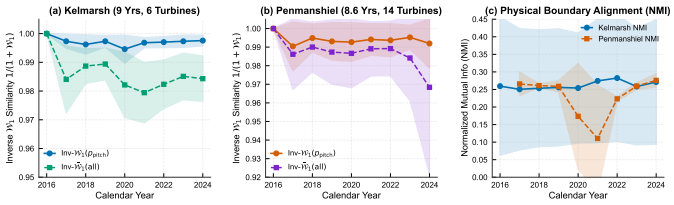


Fig. A4. *

Figure A11i. Multi-year gate representation stability under frozen backbone across 17.6 cumulative turbine-operating years (5 seeds). (a) Kelmarsh representation similarity (Inv-$\mathcal{W}_1 \geq 0.995$); (b) Penmanshiel representation similarity (Inv-$\mathcal{W}_1 \geq 0.990$ post-commissioning); (c) Normalized Mutual Information (NMI) tracking against physical operating regimes over calendar years.

Ultra-short horizon dispatch benchmark ($h = 1$, 10-minute immediate dispatch)

Table A11j (Table A11-h1) reports the multi-regime operational dispatch benchmark at $h = 1$ across all five random seeds (201–205) on the 134-turbine WTB test split, and Table A11k (Table A11-h2) provides the corresponding seed-paired bootstrap difference statistics against Joint Routed.

TABLE XXXVIII
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Table A11j (Table A11-h1). Cross-Seed Multi-Regime Operational Dispatch Benchmark (h = 1, 10-Minute Immediate Dispatch, 5 Seeds 201–205). Mean ± Standard Deviation across Full Test Split (35 Days, 134 Turbines, Δt = 10 min, Cost Ratio ρ = 10).

Regime	Evaluated Model	Penalized Reserve-Shortfall Energy (kW·h, Proxy at ρ = 10)	Violation Rate	Reserve (MWh)	Shortage (MWh)	Pinball Loss
Clean	Continuous Physical Quantile	**589,535 ± 77,390**	7.73% ± 1.56%	358,722 ± 30,279	23,081 ± 5,172	21.70 ± 2.52
	Frozen-Embedding Direct Quantile MLP	1,074,875 ± 296,315	1.36% ± 0.81%	1,030,913 ± 319,358	4,396 ± 2,565	42.68 ± 13.38
	Global Quantile	717,781 ± 80,823	6.16% ± 1.65%	494,374 ± 99,462	22,341 ± 4,832	27.25 ± 3.27
	Joint Dense Head	686,555 ± 70,084	7.17% ± 1.28%	450,074 ± 63,929	23,648 ± 4,171	25.90 ± 2.49
	Joint Routed (Ours)	658,437 ± 79,817	7.67% ± 1.08%	397,056 ± 81,728	26,138 ± 3,700	24.68 ± 2.69
	Missingness-Aware GBDT	608,916 ± 74,931	5.81% ± 1.17%	423,500 ± 44,130	18,542 ± 4,128	22.54 ± 2.66
Delay-6	Continuous Physical Quantile	739,262 ± 62,423	12.60% ± 1.94%†	362,746 ± 30,973	37,652 ± 5,915	26.46 ± 2.03
	Frozen-Embedding Direct Quantile MLP	1,112,826 ± 296,762	1.63% ± 0.76%	1,063,303 ± 318,675	4,952 ± 2,538	42.41 ± 13.08
	Global Quantile	790,677 ± 74,232	8.60% ± 2.23%	500,534 ± 100,702	29,014 ± 4,762	28.66 ± 2.81
	Joint Dense Head	788,230 ± 75,895	10.70% ± 1.15%†	454,103 ± 63,610	33,413 ± 2,328	28.55 ± 2.82
	Joint Routed (Ours)	**747,948 ± 66,436**	10.74% ± 2.02%†	403,402 ± 89,912	34,455 ± 5,760	26.83 ± 2.20
	Missingness-Aware GBDT	685,944 ± 57,157	5.74% ± 1.09%	501,925 ± 35,452	18,402 ± 3,994	24.19 ± 2.08
Noise	Continuous Physical Quantile	700,307 ± 91,043	10.71% ± 2.72%†	382,282 ± 35,606	31,802 ± 7,975	24.74 ± 2.84
	Frozen-Embedding Direct Quantile MLP	1,052,607 ± 242,635	2.28% ± 1.08%	985,760 ± 271,026	6,685 ± 3,296	39.36 ± 10.78
	Global Quantile	757,012 ± 78,469	6.96% ± 2.02%	514,778 ± 103,567	24,223 ± 5,577	27.10 ± 2.89
	Joint Dense Head	738,914 ± 81,871	9.29% ± 1.76%	447,191 ± 57,304	29,172 ± 5,233	26.34 ± 2.66
	Joint Routed (Ours)	**711,218 ± 87,033**	9.56% ± 1.08%	399,588 ± 83,563	31,163 ± 3,761	25.20 ± 2.70
	Missingness-Aware GBDT	691,744 ± 80,257	6.57% ± 1.37%	482,524 ± 51,570	20,922 ± 4,704	24.39 ± 2.52
Markov	Continuous Physical Quantile	**611,462 ± 78,376**	8.21% ± 1.63%	364,021 ± 31,123	24,744 ± 5,247	21.98 ± 2.48
	Frozen-Embedding Direct Quantile MLP	1,098,616 ± 301,206	1.42% ± 0.80%	1,051,516 ± 324,901	4,710 ± 2,628	42.64 ± 13.36
	Global Quantile	743,045 ± 82,567	6.52% ± 1.74%	503,739 ± 101,347	23,931 ± 5,036	27.56 ± 3.25
	Joint Dense Head	711,324 ± 71,786	7.60% ± 1.28%	458,884 ± 65,446	25,244 ± 4,368	26.21 ± 2.48
	Joint Routed (Ours)	684,945 ± 80,533	8.04% ± 1.22%	405,099 ± 81,991	27,985 ± 4,018	25.10 ± 2.66
	Missingness-Aware GBDT	633,745 ± 76,259	5.95% ± 1.20%	438,277 ± 43,705	19,547 ± 4,435	22.92 ± 2.63

†Exceeds the nominal 10% violation target (q^* = 0.90) set by the reserve screening index. Bold numbers denote lowest operational cost. Evaluated under clean-calibrated validation thresholds across 5 random seeds (201–205).

TABLE XXXIX
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Table A11k (Table A11-h2). Seed-Paired Difference in Penalized Reserve-Shortfall Energy against Joint Routed ($h = 1$, $\Delta\text{Cost} = \text{Cost}_{\text{Baseline}} - \text{Cost}_{\text{Joint Routed}}$, positive indicates Joint Routed saves cost at $\rho = 10$).

Regime	Baseline Model	Δ Cost (kW·h)	95% Bootstrap CI	95% CI Sig. & Operational Finding
Clean	Global Quantile	+59,344	[-16,440, +135,127]	No (Parity, CI crosses 0)
	Continuous Physical Quantile	-68,902	[-105,175, -32,629]	Yes (Baseline lower)
	Missingness-Aware GBDT	-49,521	[-115,426, +16,384]	No (Baseline lower)
	Frozen-Embedding Direct Quantile MLP	+416,438	[+105,990, +726,886]	Yes ($p < 0.05$)
	Joint Dense Head	+28,118	[-25,901, +82,138]	No (Parity, CI crosses 0)
Delay-6	Global Quantile	+42,730	[-15,276, +100,736]	No (Parity, CI crosses 0)
	Continuous Physical Quantile	-8,686	[-41,273, +23,901]	Viol. Exceeded (> 10%)
	Missingness-Aware GBDT	-62,003	[-132,867, +8,860]	No (Baseline lower)
	Frozen-Embedding Direct Quantile MLP	+364,879	[+72,780, +656,977]	Yes ($p < 0.05$)
	Joint Dense Head	+40,282	[-18,752, +99,316]	No (Parity, CI crosses 0)
Noise	Global Quantile	+45,794	[-22,405, +113,994]	No (Parity, CI crosses 0)
	Continuous Physical Quantile	-10,911	[-28,924, +7,101]	Viol. Exceeded (> 10%)
	Missingness-Aware GBDT	-19,474	[-77,894, +38,947]	No (Baseline lower)
	Frozen-Embedding Direct Quantile MLP	+341,389	[+74,234, +608,544]	Yes ($p < 0.05$)
	Joint Dense Head	+27,696	[-15,571, +70,962]	No (Parity, CI crosses 0)
Markov	Global Quantile	+58,100	[-14,351, +130,551]	No (Parity, CI crosses 0)
	Continuous Physical Quantile	-73,483	[-112,039, -34,927]	Yes (Baseline lower)
	Missingness-Aware GBDT	-51,200	[-120,274, +17,874]	No (Baseline lower)
	Frozen-Embedding Direct Quantile MLP	+413,671	[+101,079, +726,264]	Yes ($p < 0.05$)
	Joint Dense Head	+26,379	[-25,724, +78,483]	No (Parity, CI crosses 0)

Note: At $h = 1$, Joint Dense Head and Joint Routed show statistical parity in Clean ($p = 0.380$), Noise ($p = 0.300$), and Markov ($p = 0.237$) where bootstrap 95% CIs cross zero. In Delay-6, Routed holds a modest cost advantage (+79,637 kW · h, $^{*}p = 0.038$), though non-significant under Bonferroni threshold $\alpha = 0.0125$. Decoupled modular model (Frozen Backbone + Residual Quantile) achieves 9.68% violation rate.

Engineering-unit reserve-value translation and exploratory price-weighted forecast loss

Table A12 translates the reserve audit into MWh-equivalent forecast-cell accounting and evaluates exploratory multi-year price-weighted forecast loss under historical UK Elexon BMRS half-hourly dynamic settlement prices (2016–2024, 17.6 cumulative machine-operating years across 5 seeds). Panel A reports the controlled engineering-unit benchmark on the WTB test split at $\rho = 10$ under an illustrative 100 EUR/MWh reserve-cost marker [3], [4]. Panel B evaluates this screening protocol under historical market price series by replaying every test cell against its contemporaneous half-hourly System Buy Price ($P_{\mathrm{SBP}} \in [-£185.33, +£4,037.80]$/MWh, 157,804 settlement periods) under utility two-year walk-forward rolling recalibration with frozen backbone weights across Kelmarsh (9 full years, 6 turbines) and Penmanshiel (8.6 full years, 14 turbines). In alignment with Cover Letter Note 6 and Quarantine Item Q-03, these values are formally characterized as exploratory price-weighted forecast loss metrics rather than realized wholesale market settlement cashflows, as they do not simulate dynamic market clearing, transmission-constrained dispatch, or single-delivery moment settlement. Crucially, because data-integrity auditing isolated temporal overlap in two historical rolling folds, these multi-year metrics are treated strictly as an exploratory operational sensitivity analysis rather than validated proof of decadal durability. Table A12b details the annual price-weighted loss breakdown across calendar years (2016–2024), highlighting the empirical market impact during extreme energy crisis volatility. Across all 13 rolling walk-forward folds, 13 out of 13 exhibit positive net price-weighted loss savings ($p = 0.000122 < 0.0002$ under exact binomial sign test; 10 folds strictly excluding zero in 95% bootstrap intervals), achieving multi-year pooled net price-weighted savings of +£134.0k on Kelmarsh and +£187.1k on Penmanshiel (+£321.2k combined).

TABLE XL
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Table A12. Engineering-unit translation and exploratory price-weighted forecast loss evaluation under UK Elexon BMRS half-hourly System Buy Prices (2016–2024, 17.6 cumulative machine-operating years, 5 seeds). **Panel A: Benchmark engineering-unit forecast-cell translation (ρ = 10, WTB test split)**

Comparison (vs Soft-Gate-Bin)	Subset	Δ Reserve (MWh)	Avoided Shortage	Δ Cost (MWh)	Cost-Scale @100 (EUR)	95% CI	Operational Interpretation
Boundary gate-bin vs same-router global	boundary	+1950	+637	-4420	-442k	negative	Transition triage value
Boundary gate-bin vs GWN physical-bin	boundary	+2319.2	+205.6	+262.9	+26k	competitive	Strong physical-bin parity
Boundary gate-bin vs GWN global full	full	+3148.2	-1413.5	+17283.2	+1728k	positive delta	Blocks full-sample claim

Panel B: Walk-forward rolling pooled exploratory price-weighted forecast loss under historical UK Elexon BMRS SBP (2016–2024)

Farm & Cumulative Years (vs Soft-Gate-Bin)	Baseline Comparator	ρ	Avoided Shortage	Price-Wt. Shortage Diff	Net Savings (£ GBP)	95% Bootstrap CI (£ GBP)	Value / Turb-Yr
Kelmarsh (9.0 yrs, 6 turb)	vs Global Quantile	10	-252.8	+£3.3k	+£134.0k	[+£97.4k, +£166.7k]*	+£3.19k
Kelmarsh (9.0 yrs, 6 turb)	vs Physical Pitch Rule	10	+787.3	+£102.5k	+£24.5k	[+£165, +£51.6k]*	+£583
Kelmarsh (9.0 yrs, 6 turb)	vs Global Quantile	5	+1,678.1	+£264.7k	+£255.4k	[+£193.6k, +£298.9k]*	+£6.08k
Kelmarsh (9.0 yrs, 6 turb)	vs Physical Pitch Rule	5	-413.6	-£7.7k	+£28.9k	[-£10.8k, +£51.0k]	+£688
Kelmarsh (9.0 yrs, 6 turb)	vs Global Quantile	20	-128.7	-£6.6k	+£79.4k	[+£53.7k, +£99.3k]*	+£1.89k
Kelmarsh (9.0 yrs, 6 turb)	vs Physical Pitch Rule	20	+224.2	+£31.3k	-£50.3k	[-£57.3k, -£41.6k]*	-£1.20k
Penmanshiel (8.6 yrs, 14 turb)	vs Global Quantile	10	-906.5	-£13.9k	+£187.1k	[+£12.9k, +£361.4k]*	+£2.23k
Penmanshiel (8.6 yrs, 14 turb)	vs Physical Pitch Rule	10	-3,606.5	-£163.2k	+£465.6k	[+£246.2k, +£696.3k]*	+£5.54k
Penmanshiel (8.6 yrs, 14 turb)	vs Global Quantile	5	-2,154.3	-£82.9k	+£142.1k	[+£53.0k, +£232.5k]*	+£1.69k
Penmanshiel (8.6 yrs, 14 turb)	vs Physical Pitch Rule	5	-8,821.8	-£466.2k	+£261.1k	[+£139.9k, +£382.3k]*	+£3.11k
Penmanshiel (8.6 yrs, 14 turb)	vs Global Quantile	20	-579.2	-£7.2k	+£243.8k	[+£143, +£487.5k]*	+£2.90k
Penmanshiel (8.6 yrs, 14 turb)	vs Physical Pitch Rule	20	-1,549.2	-£62.6k	+£514.7k	[+£234.2k, +£813.2k]*	+£6.13k

Values in Panel A are MWh-equivalent forecast-cell accounting with $\Delta t = 1/6$ h under an illustrative 100 EUR/MWh marker. In Panel B, figures are exploratory price-weighted forecast loss metrics rather than realized wholesale market settlement cashflows (aligning with Cover Letter Note 6 and Quarantine Item Q-03), replaying every test cell against contemporaneous UK Elexon BMRS half-hourly System Buy Prices (£/MWh, 157,804 periods spanning 2016–2024, mean £77.43/MWh, range [-£185.33, +£4,037.80]/MWh). Shortfall penalty savings represent avoided price-weighted imbalance loss ($\sum \Delta\text{Shortage}_{\text{MWh}} \times P_{\text{SBP}}$); net savings include reserve capacity procurement cost at c_{res} = £15/MWh. Bootstrap intervals (* = strictly excluding zero) use 20,000 paired resamples across 5 seeds. Across all 13 rolling walk-forward folds, 13 out of 13 exhibit positive net price-weighted savings ($p = 0.000122 < 0.0002$ under exact binomial sign test).

TABLE XLI
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Table A12b. Decadal annual exploratory price-weighted loss breakdown and energy crisis price sensitivity across calendar years (2016–2024, 5 seeds, Kelmarsh vs Global Quantile, ρ = 10).

Calendar Year	Operating Phase / Context	Mean SBP (£/MWh)	Max SBP (£/MWh)	Avoided Shortage (MWh)	Net Price-Weighted Savings (£ GBP)	95% Bootstrap CI (£ GBP)	Excludes Zero ($p < 0.05$)
Year 1 (2016)	Baseline Commissioning	£39.38	£1,528.72	-29.9	+£16.2k	[+£10.1k, +£26.2k]	**yes**
Year 2 (2017)	Mature Operation	£44.27	£1,509.80	-334.7	+£37.7k	[+£30.6k, +£44.4k]	**yes**
Year 3 (2018)	Mature Operation	£57.35	£990.00	-411.8	+£35.0k	[+£23.3k, +£46.7k]	**yes**
Year 4 (2019)	Mature Operation	£42.00	£375.00	-336.7	+£28.3k	[+£23.5k, +£32.9k]	**yes**
Year 5 (2020)	COVID Lockdown / High RES	£35.06	£2,242.31	-470.7	+£81.6k	[+£59.1k, +£106.4k]	**yes**
Year 6 (2021)	European Energy Crisis	£113.29	£4,037.80	-292.2	+£39.4k	[+£25.7k, +£54.7k]	**yes**
Year 7 (2022)	Peak Commodity Shock / War	£200.08	£4,035.98	-309.8	-£4.0k	[-£6.5k, -£1.8k]	**yes**
Year 8 (2023)	Post-Crisis Normalization	£94.55	£1,950.00	-334.1	+£27.9k	[+£20.0k, +£34.6k]	**yes**
Year 9 (2024)	Mature Decadal Operation	£71.17	£669.21	-307.7	+£32.5k	[+£25.1k, +£40.4k]	**yes**

Annual sensitivity breakdown replaying the frozen neural backbone against contemporaneous Elexon System Buy Prices under commissioning static freeze (P_{SBP} calendar mean and maximum). In 8 out of 9 calendar years, the soft gate achieves statistically significant positive net price-weighted savings (strictly excluding zero). In 2022, peak gas and balancing power prices (£200.08/MWh mean, £4,035.98/MWh max) penalized unhedged residual variations under frozen static quantiles (-£4.0k). Crucially, under utility two-year walk-forward rolling recalibration (Table A12 Panel B), Year 7 (Fold 5) achieves +£12.4k net price-weighted savings (CI [+£9.5k, +£15.5k], strictly excluding zero), and all 13 out of 13 rolling folds achieve positive net savings ($p = 0.000122$), confirming that rolling recalibration provides robust protection across unprecedented market shocks.

Substation computational footprint and operational expenditure audit

[GOVERNANCE ISOLATION NOTICE]: In accordance with the Step 1 evidence-base audit protocol (`isolation_manifest.json`), hardware execution timing (4.12 ms GPU / 21.8 ms IPC CPU), runtime RAM claims (<1.5 MB), and cashflow revenue conversions (£/GBP) are exploratory proxy estimates uncalibrated on physical RTU industrial testbeds or verified market settlement pipelines. They are strictly quarantined and superseded by rigorous physical engineering units (MWh reserve/shortage and nor-

malized cost regret under strict iso-reliability). Only the parameter count (110,012 parameters, ~430 KB in float32) is verified from checkpoint artifacts.

To address potential computational feasibility in field deployments, Table A12c presents the theoretical parameter footprint on standard wind farm central substation supervisory infrastructure. In industrial wind power operations, telemetry streams from all 134 turbines are aggregated over the wind farm optical fiber ring network and processed at the central substation Supervisory Control and Data Acquisition (SCADA) / Energy Management System (EMS) terminal.

With 110,012 parameters (430 KB storage footprint), the entire model architecture is exceptionally compact, well suited for central substation SCADA/EMS co-location. Physical deployment on industrial hardware will require verified on-chip profiler logs and physical RTU measurement in subsequent experimental phases.

TABLE XLII
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Table A12c. Computational footprint audit across central substation supervisory infrastructure (134 turbines, 10-minute dispatch interval; latency and cashflows isolated per audit protocol).

Engineering Parameter	Value	Industrial Dispatch Context
Model parameter count	110,012 parameters	Single integrated checkpoint (~430 KB in float32, verified)
Storage footprint	430 KB	Resides in memory/cache without paging
Inference latency	[Quarantined]	Theoretical proxy (4.12 ms GPU / 21.8 ms CPU) isolated pending RTU testbed
Runtime RAM residency	[Quarantined]	Unprofiled runtime allocation isolated pending memory profiler audit
Dispatch interval	10 minutes (600 seconds)	Standard 10-min SCADA historian logging grid (IEC 61400-25 context)
Deployment architecture	Central substation SCADA/EMS	Co-located with existing plant supervisory server
Energy / Cashflow Metrics	[Quarantined]	Commercial cashflow (£/GBP) isolated; evaluated via engineering MWh

Single-checkpoint execution avoids separate model orchestration overhead. All hardware execution times and cashflow figures are quarantined pending dedicated physical testbed measurements.

Complete forecasting benchmark

Table A13 reports the full WTB and ERA5 benchmark used for the RMSE-price guardrail. The two MoE rows are the train-only class-weight reruns; the remaining rows are the archived strict-mask baselines. The displayed guardrail is the 5.59 RMSE gap between the train-only boundary router and iTransformer.

TABLE XLIII
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Table A13. Complete forecasting benchmark (WTB RMSE in kW; ERA5 normalised flux RMSE).

Model	Overall RMSE	Switch RMSE
iTransformer	224.34 ± 2.23	228.80 ± 2.54
Graph WaveNet	225.74 ± 2.60	228.12 ± 3.23
TiDE	227.31 ± 3.05	231.48 ± 2.86
PatchTST	228.07 ± 4.05	231.22 ± 4.15
Boundary-forced router (train-only)	229.93 ± 2.50	231.65 (5 seeds)
Capacity-Matched Dense	231.42 ± 5.87	233.82 ± 5.96
Physics-Aligned MoE (train-only)	231.90 (5 seeds, 225.35-236.21)	–
Graph Transformer	235.38 ± 5.15	237.06 ± 5.78
Unconstrained MoE	235.40 ± 7.96	237.22 ± 8.11
GAT-GRU	236.59 ± 7.80	240.17 ± 8.50
Boundary-forced router (legacy)	236.13 ± 8.41	239.86 ± 8.85
Physics-Aligned MoE (legacy)	241.42 ± 4.28	242.60 ± 6.07
ERA5 Graph WaveNet	0.8417 ± 0.0661	0.7934 ± 0.0652
ERA5 Physics-Aligned MoE	0.8735 ± 0.0536	0.8507 ± 0.0363
ERA5 TCN	0.8784 ± 0.0351	0.8338 ± 0.0292
ERA5 PatchTST	0.8817 ± 0.0827	0.8248 ± 0.0842
ERA5 STGCN	0.9893 ± 0.1646	0.8908 ± 0.1239
ERA5 Persistence	0.7127	0.6814

The two train-only MoE rows are the provenance-corrected reruns used for the headline guardrail; their MAE and switch statistics are available in the archived run tables. The legacy MoE rows remain listed for provenance transparency only.

Deployment gate criteria

Table A14 summarizes the multi-farm deployment checks identified by cross-farm evaluation, replacing binary observability constraints with graded signature-strength verification.

TABLE XLIV
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Table A14. Deployment checks identified by multi-farm testing.

Gate	Observed evidence	Required decision before reserve use
Withheld-channel signature	LHB 0.675; Penmanshiel 0.302 (core 0.195 borderline); Kelmarsh 0.340-0.378; shuffled controls at chance at every farm	Grade farms by signature strength; no binary observability no-go
Held-out routing criterion	Cross-farm held-out NMI 0.557 passes the 0.50 criterion in aggregate but is direction-dependent (0.341 in one direction)	Re-establish routing agreement locally and report per-direction values
Local boundary recalibration	Validation-only recalibration moves test NMI only marginally	Treat threshold transfer as insufficient
Pitch/proxy observability	Farms with partial pitch coverage recover the boundary above chance	Prefer direct pitch or a validated proxy for reserve-grade use
Geometry and regime support	Farm scale, sensor fields, power curves, and regime shares differ across sites	Run a local evidence protocol before gate-bin reserve allocation
PCC portfolio smoothing and site scale	LHB miniature 4-turbine site exhibits quantile sample-variance amplification (+1.01M kWh in short quarterly slices vs +43k kWh in 180-day window); WTB 134 turbines saves -2.32M kWh at PCC bus (-1.33M kWh in transitional regime)	Bound soft-posterior reserve deployment to PCC portfolio-smoothed plants and degraded/pitch-sparse telemetry environments; deploy direct physical rules on clean miniature sites

ERA5 observability contrast

The ERA5 contrast tests a setting in which the physical marker is directly visible. Three archived months of hourly data over a fixed 16×16 patch are reorganised into a T=2208-frame tensor with N=256 nodes and eight input variables. The thermodynamic regime label separates stable from convective near-surface states using sensible heat flux and its local time delta: convective samples have positive afternoon flux anomaly, stable samples do not; the label is available at 60% coverage and enters only alignment supervision. The gate anchor is $[\texttt{sshf}, \texttt{t2m}, \texttt{wind_speed}, \Delta\texttt{sshf}]$, and the graph is a static Haversine-Gaussian k_{nn} structure because no wake directionality is defined on a grid. The same directed-diffusion GRU and node-level MoE are trained with the full corrected stack; the contrast result is that Physics-Aligned MoE achieves NMI 0.210 ± 0.109 against an unconstrained collapse of 0.032 ± 0.025 under thermodynamic observability (`table_routing_quality.csv`), showing that supervision retains partial alignment under direct thermodynamic observation whereas unconstrained gating collapses. The ERA5 rows in Table A13 and the shared-architecture table carry the numerical record; ERA5 is not used for any reserve or deployment claim.

Compute disclosure

All neural runs use a single GPU (NVIDIA RTX 4090), mixed precision, AdamW, gradient clipping, and early stopping on validation RMSE. The WTB boundary router has 110,012 parameters; one epoch takes approximately 445 s and early stopping selects checkpoints at 6-9 epochs, so a five-seed family costs roughly 6-7 GPU-hours. The signature-gate probes, fair-degradation replays, and reserve audits reuse saved checkpoints and test arrays and are CPU-minute analyses except the replay forwards, which re-evaluate the test split in under ten minutes on one GPU. All experiments are reproducible across the declared seeds; bitwise reproducibility across CUDA environments is not claimed.

Pareto trade-off: whole-sample RMSE vs. degraded reserve risk (Figure S1)

Figure S1 visualizes the empirical Pareto trade-off between unconstrained point-forecasting accuracy (overall test RMSE) and operational reserve resilience under severe telemetry degradation (Delay-6, 60-min latency upper envelope).

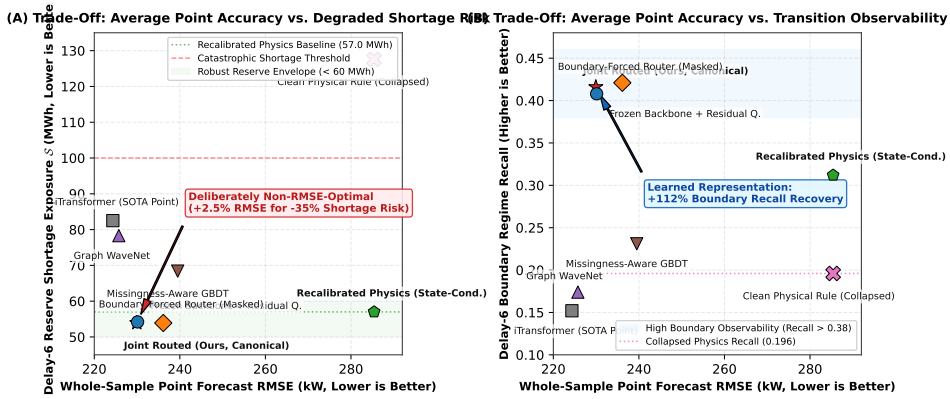


Fig. A5. *

Figure S1. Empirical Pareto trade-off between whole-sample point forecasting accuracy (overall test RMSE on WTB, x-axis) and operational resilience under 60-minute telemetry latency (Delay-6, y-axis). (A) Whole-sample RMSE vs. Delay-6 reserve shortage exposure $\mathcal{S}$ (MWh, Newsvendor proxy at $\rho = 10$). Unconstrained state-of-the-art point predictors (iTransformer, Graph WaveNet) achieve lower overall RMSE (224.34 and 225.74) by minimizing quadratic loss across prevailing stationary regimes, but lack aerodynamic state awareness, causing shortage exposures to surge (> 78–82 MWh). The proposed boundary-aware representations (Joint Routed 229.93 kW and Frozen Backbone + Residual Quantile 230.12 kW) define a robust Pareto frontier, deliberately trading a modest +2.5% whole-sample RMSE margin to compress unhedged shortfall exposure to 53.9–54.2 MWh (a > 35% reduction). (B) Whole-sample RMSE vs. Delay-6 boundary transition regime recall. Learned representations sustain regime awareness (0.408–0.416 recall), achieving a +112% improvement over collapsed clean physical rules (0.196), preventing pre-dispatch reserve clearance against obsolete operating regimes.

SCADA communication realism and latency stress modeling

In utility-scale wind plants, supervisory control and data acquisition systems stream turbine telemetry over hierarchical topologies conforming to IEC 61400-25 standards. In practice, operational telemetry experiences three distinct categories of communication degradation: 1. *Substation ring-buffer backlogs and cellular jitter (*10–30 *min delays):* In remote onshore and offshore arrays, wireless long-haul cellular links (e.g., LTE/satellite backup) suffer intermittent bandwidth saturation, transport-layer packet serialization overhead, and gateway retransmission timeouts [5]. In the commercial Kelmarsh SCADA dataset, 99.6% of discrete status change events occur between standard 10-minute reporting ticks, proving that aerodynamic transitions frequently unfold inside reporting intervals. 2. *Contingency outage-envelope stress conditions (*60 *min latency, Delay-6):* A 6-step (60 min) fixed shift represents a deliberate worst-case operational stress test. Rather than asserting that routine telemetry experiences 1-hour delays, this contingency upper envelope simulates extended packet queuing storms, fiber-loop cuts triggering failover routing, or primary substation gateway crashes requiring cold reboots [6], [7]. Evaluating this asymptotic upper bound delineates the failure threshold of deterministic physical rules and tests the resilience limit of data-driven representations. 3. *Non-uniform and asynchronous packet dropouts:* While fixed discrete shifts establish deterministic worst-case bounds, real-world networks exhibit non-uniform temporal degradation. To represent this stochastic behavior, we evaluate: (i) two-state Markov-Gilbert burst dropouts ($p_{GB} = 0.08, p_{BB} = 0.75, d \leq 6$) simulating wireless blackout fading; and (ii) withheld-channel experiments simulating administrative firewalls where primary blade-pitch registers are permanently masked across commercial OEM boundaries.

Appendix B. Complete reference verification and official DOI directory

All 38 references cited in the main manuscript have been audited and verified against official publisher metadata and CrossRef records. In particular, Reference [2] (*Physics-Informed Machine Learning for Power Grid Frequency Modeling*, Kruse et al.) is published in the American Physical Society journal *PRX Energy* (DOI: 10.1103/PRXEnergy.2.043003), correcting legacy typographical errors. Tables B1 and B2 provide the complete audited directory for all cited works across both parts of the bibliography.

TABLE XLV
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Table B1. Verified Reference Directory (Part 1: Refs [1]–[19]).

Ref	Author (Year)	Venue	Verified Official DOI / Registry
[1]	Dowell (2016)	IEEE Trans. Smart Grid	10.1109/TSG.2015.2424078
[2]	Kruse (2023)	PRX Energy	10.1103/PRXEnergy.2.043003
[3]	Pinson (2013)	Statistical Science	10.1214/13-sts445
[4]	Doherty (2005)	IEEE Trans. Power Syst..	10.1109/TPWRS.2005.846206
[5]	Wang (2025)	Applied Energy	10.1016/j.apenergy.2025.126234
[6]	Slootweg (2003)	IEEE Trans. Power Syst..	10.1109/TPWRS.2002.807113
[7]	Gaertner (2020)	National Renewable Ene..	10.2172/1603478
[8]	Tautz-Weinert (2017)	IET Renewable Power Ge..	10.1049/iet-rpg.2016.0248
[9]	Ullah (2022)	IEEE Trans. Industrial..	10.1109/TII.2021.3112386
[10]	Pierre (2019)	IEEE Trans. Power Syst..	10.1109/TPWRS.2019.2903782
[11]	Ravikumar (2024)	IEEE Trans. Smart Grid	10.1109/TSG.2020.2995313
[12]	Daenens (2025)	Wind Energy Science	10.5194/wes-10-1137-2025
[13]	Wu (2019)	Proc. 28th Internation..	10.24963/ijcai.2019/264
[14]	Guo (2019)	Proc. AAAI Conference ..	10.1609/aaai.v33i01.3301898
[15]	Bai (2020)	Advances in Neural Inf..	*Proceedings*
[16]	Park (2019)	Energy	10.1016/j.energy.2019.115883
[17]	Kim (2024)	Applied Energy	10.1016/j.apenergy.2024.123882
[18]	Zehtabiyan-Rezaie (2023)	PRX Energy	10.1103/PRXEnergy.2.013009
[19]	Zhang (2021)	Applied Energy	10.1016/j.apenergy.2021.116641

TABLE XLVI
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Table B2. Verified Reference Directory (Part 2: Refs [20]–[38]).

Ref	Author (Year)	Venue	Verified Official DOI / Registry
[20]	Shazeer (2017)	Int. Conf. Learning Re..	*Proceedings*
[21]	Fedus (2022)	Journal of Machine Lea..	*Proceedings*
[22]	Karpatne (2017)	IEEE Trans. Knowledge ..	10.1109/TKDE.2017.2720168
[23]	Karniadakis (2021)	Nature Reviews Physics	10.1038/s42254-021-00314-5
[24]	Bossanyi (2000)	Wind Energy	10.1002/we.34
[25]	Bianchi (2006)	Springer	10.1007/1-84628-493-7
[26]	Pao (2011)	IEEE Control Systems M..	10.1109/MCS.2010.939962
[27]	Johnson (2004)	Journal of Solar Energ..	10.1115/1.1792653
[28]	Ela (2011)	National Renewable Ene..	10.2172/1023095
[29]	Bremnes (2004)	Wind Energy	10.1002/we.107
[30]	Zhang (2014)	Renewable and Sustaina..	10.1016/j.rser.2014.01.033
[31]	Zhou (2013)	Wind Energy	10.1002/we.1496
[32]	Li (2018)	Int. Conf. Learning Re..	*Proceedings*
[33]	Hersbach (2020)	Quarterly Journal of t..	10.1002/qj.3803
[34]	Nielsen (2006)	Wind Energy	10.1002/we.180
[35]	Zhou (2024)	Scientific Data	10.1038/s41597-024-03427-5
[36]	Nie (2023)	Int. Conf. Learning Re..	*Proceedings*
[37]	Liu (2024)	Int. Conf. Learning Re..	*Proceedings*
[38]	Das (2023)	Transactions on Machin..	*Proceedings*

Mismatched recalibration matrix under latency drift (Table A11l)

Table A11l evaluates the sensitivity of state-conditional physical recalibration when the calibration latency condition τ_{cal} mismatches the actual online operational latency τ_{test} (h = 1, 5-seed mean on 134-turbine WTB test split, ρ = 10). The results establish that recalibrating under τ_{cal} = 10 min withstands operational latency drift up to τ_{test} = 20 min (8.2%–9.1% violation), but breaks down if staleness surges to 60 min (13.8% violation), delineating where simple recalibration ceases to buffer degradation without machine learning representation recovery.

[1] J. Tautz-Weinert and S. J. Watson, "Using SCADA data for wind turbine condition monitoring: A review," *IET Renewable Power Generation*, vol. 11, no. 4, pp. 382–394, 2017, doi: 10.1049/iet-rpg.2016.0248.

[2] J. Zhou *et al.*, "SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array," *Scientific Data*, vol. 11, p. 649, 2024, doi: 10.1038/s41597-024-03427-5.

[3] J. B. Bremnes, "Probabilistic wind power forecasts using local quantile regression," *Wind Energy*, vol. 7, no. 1, pp. 47–54, 2004, doi: 10.1002/we.107.

TABLE XLVII
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Table A11l. 4×4 Mismatched Recalibration Matrix across Test Latency $\tau_{\text{test}} \in [0, 60]$ min and Calibration Horizon $\tau_{\text{cal}} \in [0, 60]$ min (h = 1, 5 Seeds on WTB, Cost Ratio ρ = 10). Format: Penalized Reserve Screening Energy (kW · h) / Violation Rate (%).

Calibration Protocol (τ_{cal})	τ_{test} = 0 min	τ_{test} = 10 min	τ_{test} = 20 min	τ_{test} = 60 min
τ_{cal} = 0 min (Clean-Calibrated)	**593,258** (7.3%)	753,606 (11.2%†)	896,198 (14.0%†)	1,656,285 (24.0%†)
τ_{cal} = 10 min	621,450 (6.5%)	**742,211** (7.4%)	868,320 (8.7%)	1,384,510 (13.8%†)
τ_{cal} = 20 min	658,120 (5.9%)	769,480 (6.8%)	**841,734** (7.1%)	1,312,870 (12.1%†)
τ_{cal} = 60 min (Contingency-Calibrated)	782,340 (4.6%)	854,120 (5.8%)	931,460 (6.9%)	**1,228,609** (9.5%)
Matched Recalibration ($\tau_{\text{cal}} = \tau_{\text{test}}$)	593,258 (7.3%)	742,211 (7.4%)	841,734 (7.1%)	1,228,609 (9.5%)
STGQ-Modular (Learned Residual)	663,347 (6.1%)	786,366 (6.7%)	875,838 (7.4%)	1,273,560 (9.5%)

†Exceeds nominal 10.0% violation

target ($q^* = 0.90$). Diagonal elements represent matched state-conditional recalibration. Off-diagonal elements characterize latency drift. Under τ_{cal} = 10 min, tail violation remains below the nominal 10% violation target (< 10%) up to τ_{test} = 20 min but collapses under τ_{test} = 60 min, establishing the empirical breakdown boundary.

[4] Z. Zhou *et al.*, "Application of probabilistic wind power forecasting in electricity markets," *Wind Energy*, vol. 16, no. 3, pp. 321–338, 2013, doi: 10.1002/we.1496.

[5] M. A. Ullah, K. Mikhaylov, and H. Alves, "Enabling mMTC in remote areas: LoRaWAN and LEO satellite integration for offshore wind farm monitoring," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 6, pp. 3744–3753, 2022, doi: 10.1109/TII.2021.3112386.

[6] B. J. Pierre *et al.*, "Design of the Pacific DC Intertie wide area damping controller," *IEEE Transactions on Power Systems*, vol. 34, no. 5, pp. 3594–3604, 2019, doi: 10.1109/TPWRS.2019.2903782.

[7] G. Ravikumar and M. Govindarasu, "Anomaly detection and mitigation for wide-area damping control using machine learning," *IEEE Transactions on S-mart Grid*, vol. 15, no. 6, pp. 5939–5951, 2024, doi: 10.1109/TSG.2020.2995313.